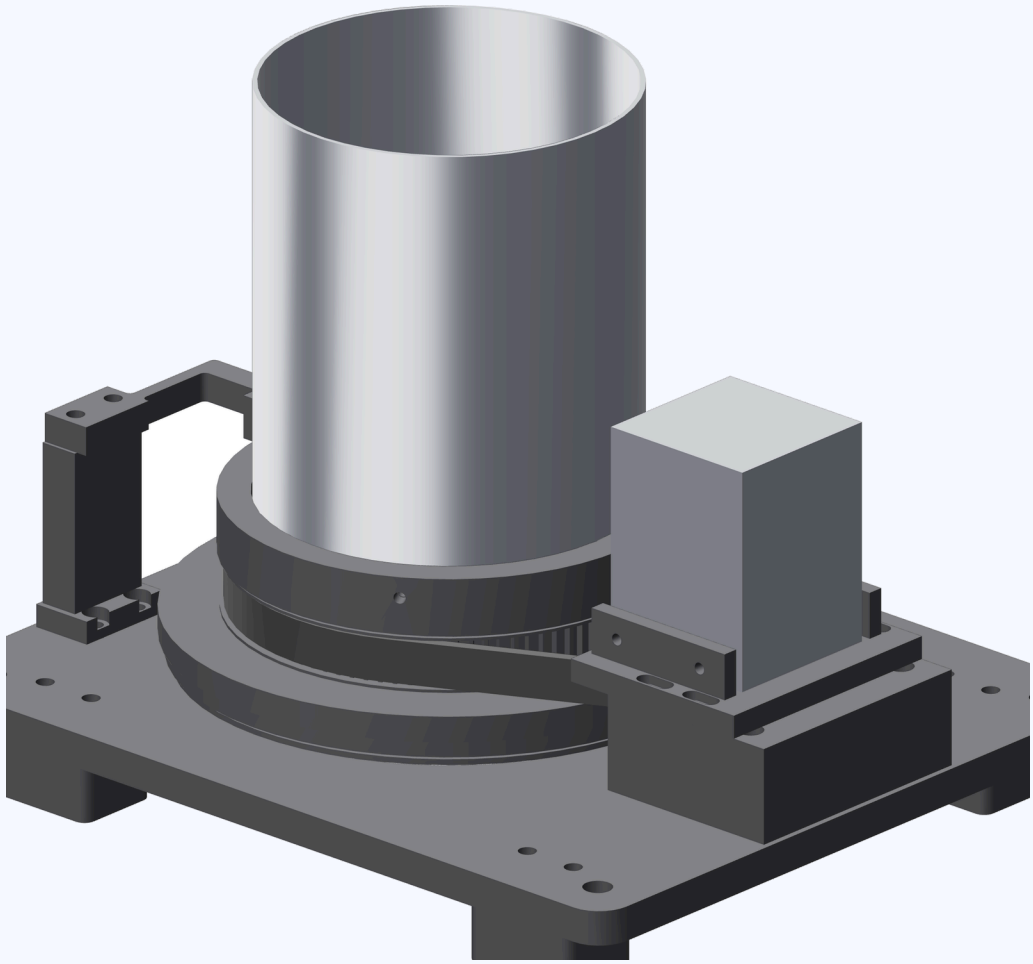
HOME SODA MACHINE

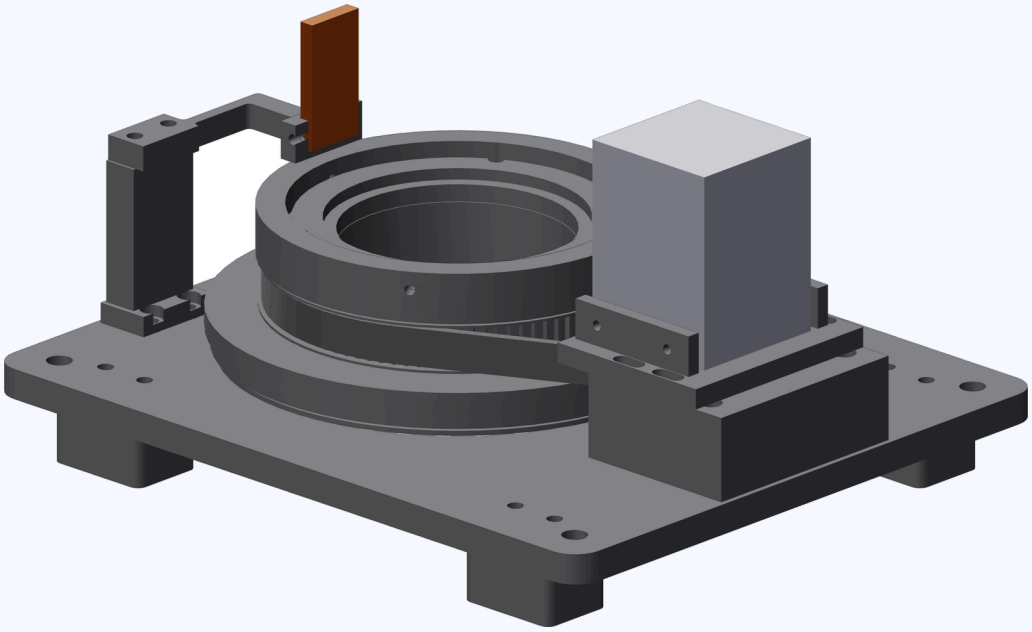
Weld rotator

Build guide for the bench fixture that turns the 5-inch carbonator tube under the laser head. Print to finished, indicated, commissioned rig — every step.



What you are building

A turntable that stands the carbonator tube vertically and rotates it, slowly and evenly, while you hold the welding head still and close an end cap.



The rig with no tube in it. A 300 × 250 mm printed base on four feet, a ball-race turntable carrying an integral 90-tooth pulley, a NEMA 23 on a sliding carriage, and the copper shoe on its flexure arm at the left.

IT DOES FOUR THINGS

- Holds the tube on its own axis to under 0.25 mm of runout.
- Turns it at a bead speed you choose between 5 and 15 mm/s.
- Stops the instant your foot leaves the pedal.
- Keeps the welder's work lead on stationary copper, so no cable winds up.

IT DOES NOT

- Carry the welding head. You hold that.
- Command the laser. The pedal is rotation only.
- Replace any guarding, eye protection, extraction, or argon control.

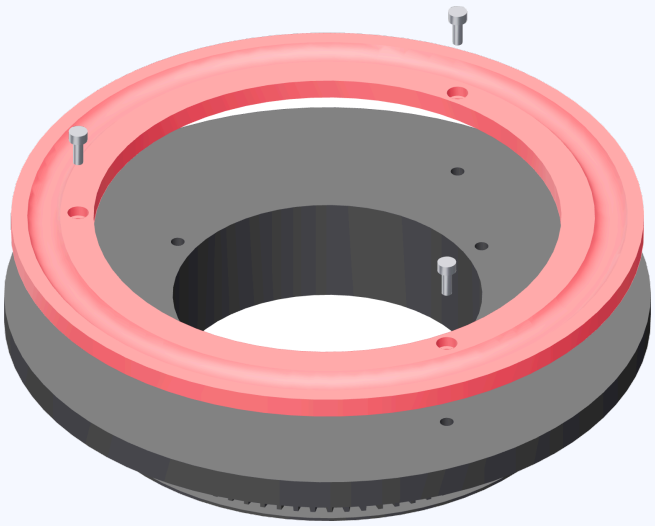
ONE NUMBER TO KEEP

3,200 pulses per motor revolution through a 4.5:1 belt is **14,400 pulses per table revolution**. The console counts them into degrees turned, but nothing counts a lap: the pedal starts and stops the table, and you end the bead at the index mark.

How to read a picture here

Every part of this fixture prints in the same black stock, so the pictures colour three things the real rig does not.

IN A PICTURE	MEANS
Coral	The part or the fastener *this step adds*. Everything already standing stays black.
Brass	A heat-set threaded insert.
Blue	A gauge, a probe, a packing block or the dial indicator — something on the bench that is not part of the fixture.
Grey metal	The motor, the pulley, a screw, or the 316L tube.



Screws stand **off** their holes, never in them.

Fasteners are drawn poised

A screw shown home is a screw you cannot see. Every fastener in this guide floats a short way off its own hole, along the axis it goes in on. The picture says *which hole and which screw*. It is not a picture of the finished joint.

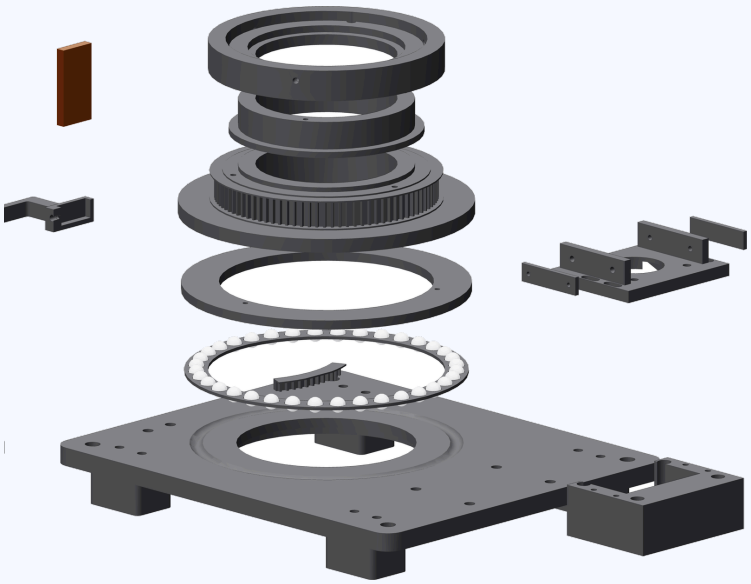
Screw and insert proxies are drawn to catalogue head and length so you can tell an M3 × 8 from an M3 × 25 at a glance. Do not measure them off the page.

A BOX LIKE THIS IS A GATE

It is the reading that says a step is finished. **A gate that does not pass is not a thing to work around.** Fix the cause and take the reading again before the next step.

The whole build

Fourteen printed shapes, one crosscut of copper bar and a controller, in nine parts and this order.



Every printed shape, lifted apart along the axis it goes together on. The copper shoe, top left, is the only piece that is not printed.

PART OF THE JOB	PAGES
Print	8–11
Heat-set inserts	12–15
Base and feet	16–17
Bearing	18–21
Drive	22–28
Nest and contact	29–32
Controller	33–39
Commissioning	40–45
Welding with it	46–47

DO NOT SKIP AHEAD

The bearing has to run free before the drive goes on, and the drive has to time correctly before a tube is loaded. Each is a gate, each is cheap to check, and each is expensive to unwind.

What you print

The printed parts use Polymaker Fiberon PET-GF15 black through the 0.4 mm nozzle. Their STEP files are in `hardware/printed-parts/fixtures/weld-rotator/`.

FILE	QTY	WHAT IT IS
base	1	The 300 × 250 × 12 mm bed. Lower ball race, bench-clamp holes, tower stations.
base-foot	4	Holds the base 24 mm off the bench so the lower ports stay reachable.
turntable-90t	1	The moving disk, with the 90-tooth pulley and the nest register on it.
race-ring	1	Upper bearing race. Printed groove-up, screwed on groove-down.
spool	1	Hub and lift-catch flange, through the base from below.
ball-cage	1	Spaces the 36 balls. Carries no load.
tube-nest	1	The replaceable register the tube end lands in. The only tube-size-specific part.
motor-tower	1	Two rails and a rear wall, all outside the belt's swept path.
motor-carriage	1	The sliding motor mount. Belt tension is its four slotted screws.
motor-clamp-pad	2	Load spreaders under the carriage's four side screws.
ground-tower	1	Stationary support for the copper contact.
ground-arm	1	The 5.0 mm leaf spring that holds the shoe on the tube. Replaceable.

In this order

1. The **base**. Nine hours; watch its first two layers. Page 9.
2. The **turntable** and the **race ring**. Page 10.
3. Everything else, in any order. Page 11.

THE COPPER SHOE IS CUT

`ground-shoe` is one 25 mm crosscut from the acquired 1/4 × 2 inch C110 copper bar. Its STEP is there so the reference assembly closes; nothing prints it and nothing drills it.

ONE PART IS TUBE-SPECIFIC

Only the `tube-nest`. A fit correction, or a later tube size, costs that one 150 mm print and leaves the base, the bearing, the pulley and the motor mount alone.

What you buy

All of it is already recorded in `hardware/ledger/purchases.md`. Check every row is in your hand before the first insert goes in.

DRIVE AND POWER

ITEM	ASIN
STEPPERONLINE 1-axis kit — NEMA 23 23HS30-2804S + DM542T driver	B089Y8H8L5
BTF-LIGHTING 24 V 4 A adapter, with its barrel-to-wire connector	B0D2XZCGL5
uxcell HTD-5M 20-tooth pulley, 15 mm belt, 6.35 mm bore	B0747KM7MN
uxcell HTD-5M belt, 110 teeth / 550 mm × 15 mm	B0CXPM3KTP
HimaPro 50122 cast-aluminium deadman pedal, SPDT	B0CYYYHL31

CONTROLLER

ESP32-DevKitC-32E	B09MQJWQN2
ULN2803A high-current driver module	B0F872W528
Chanzon 3.3 kΩ 1/4 W metal-film resistor — one of the 100	B08QRG7JBY
Universal 5 V / 3 A adapter, using its Micro-USB tip	shop stock
Micro-USB data cable	shop stock

BEARING, CONTACT, METROLOGY

uxcell PP bearing balls, 10 mm solid — 36 of the 200	B0FHJ1X6ZP
YTKavq 1/4 × 2 × 12 in C110 copper bar — one 25 mm slice	B0DR2PX6TT
Neoteck 0.0005 in dial test indicator + 40 kg magnetic base	B09W2R3SCD

HEAT-SET INSERTS

ITEM	QTY	ASIN
ruthex M5 × 9.5 brass	6	B07YSVXWS8
ruthex M3 short, RX-M3S×4.0 brass	31	B09ZHSGHXD

SCREWS

ITEM	QTY	WHERE
M5 × 20 countersunk, DIN 7991, 3 mm hex	2	B0DX1G3Y6W
M5 × 0.8 square nut, 8 across × 4 thick	2	B0F6B5X6CX
M5 × 10 socket-head cap	6	stock
M3 × 25 — feet, spool, adjusters	14	stock
M3 × 10 — carriage, nest retainers	7	stock
M3 × 8 flush — race ring	3	stock
M3 × 8 — motor pads, shoe clamp	5	stock
M3 × 12 — ground arm	2	stock

HARNESS

22 AWG wire, bootlace ferrules, 1/4 in braided sleeve, heat-shrink, WAGO 221 lever nuts	stock

TWO ROWS WORTH CHECKING TWICE

The **20-tooth pulley** went missing once and was reordered; make sure the one in your hand is the 15 mm-belt, 6.35 mm-bore part. And the copper bar is **1/4 × 2 inch**, not 1/8 — the shoe's thickness is what sets its preload.

What you need at hand

Three hex sizes cover the whole fixture. Everything else on this page is either for the inserts, the copper, or the metering.

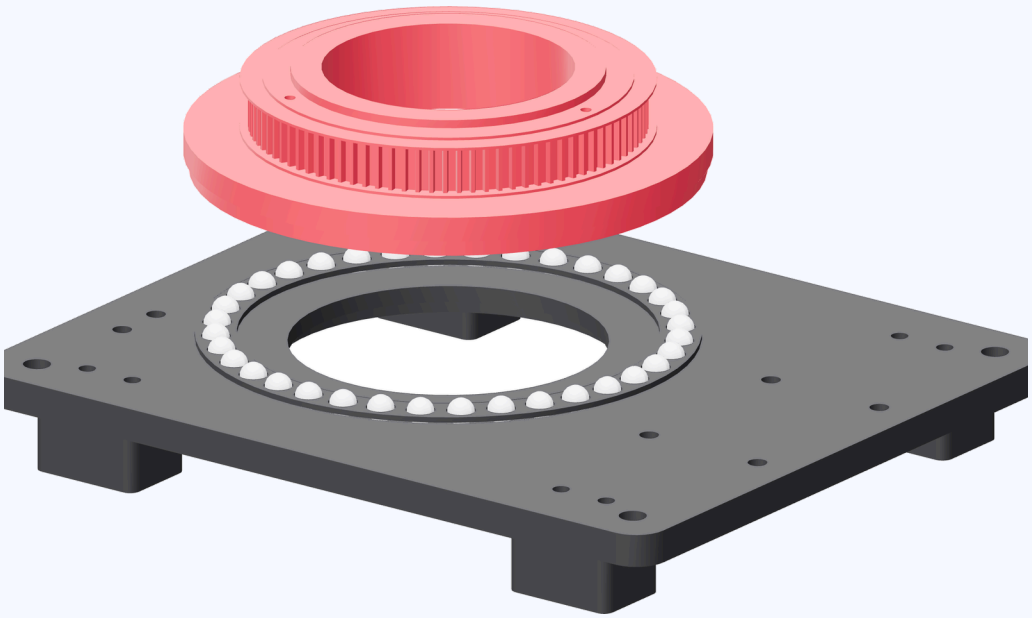
TOOL	FOR
2.5 mm hex	Every M3 in the fixture — 30 of them.
4 mm hex	The six M5 × 10 socket-head screws, tower and ground tower.
3 mm hex	The two M5 × 20 countersunk motor screws. DIN 7991 does not share the cap-screw socket.
A slim-shank 2.5 mm bit or a plain L-key	The three nest retainers. Their Ø6.2 mm access wells are narrower than a 1/4 inch bit body.
HOTO PixelDrive + M3 and M5 insert tips	All 37 heat-set inserts. Its 0.5 N·m floor is the point — see page 12.
Soldering iron with M3 / M5 insert tips	Pressing the inserts themselves.
0.25 mm / 0.010 inch feeler blade	Gauging the motor pulley off the face pilot.
Digital calliper	Sorting the 36 balls, and measuring the pulley's actual overall length.
Multimeter with continuity beep	Metering the harness before power, and the shoe-to-tube contact.
Stopwatch or a phone	Timing the dry revolutions at 5, 8 and 15 mm/s.
BA4555 bandsaw, 24 TPI M42 blade	The single 25 mm crosscut in the copper bar.
Plastic scraper	Deburring both bearing races. **Nothing abrasive.**
A 25.4 mm round bar or socket	The service-envelope probe on page 18.
Two blocks about 25 mm tall	Standing the carriage skin-up while the motor goes on.
Laptop with PlatformIO, and a Micro-USB data cable	Flashing and setting the controller, pages 37–39.
Paper index tape, marker	The direction check and the lap check.

THE ONE THING YOU WILL WANT AND NOT HAVE

The nest retainers. A normal 1/4 inch hex bit will not enter their access well. Find the slim bit or the L-key *before* the nest is on the turntable, not after.

The turntable drive

The belt connects the purchased motor pulley to the pulley printed into the turntable. The motor carriage sets their separation.



The turntable carries the driven pulley and tube nest above the bearing.

The printed side

The pulley teeth and both flanges are integral to the turntable. Its flat underside is the print datum and the mounting face for the race ring and spool.

The race ring runs on the balls in the stationary base. The spool connects beneath the turntable through the base's center.

The motor side

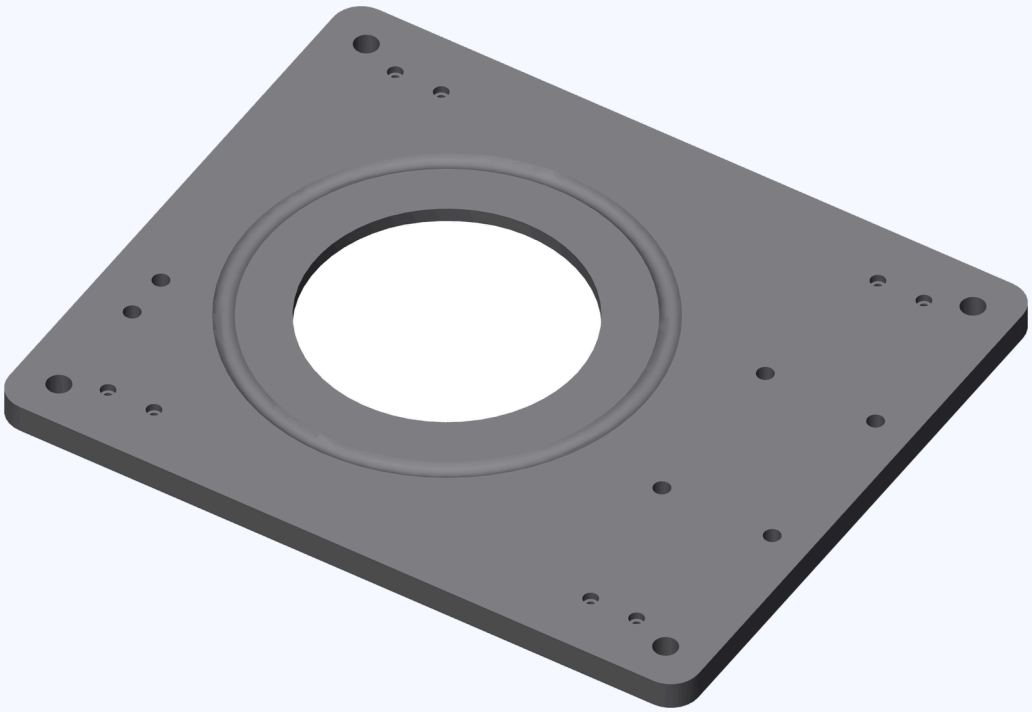
The purchased pulley sits on the motor shaft. The motor and carriage move together along the tower rails to set belt tension.

The belt runs between the pulley flanges. Its spans pass clear of the tower and carriage throughout the available adjustment.

PRINT · STEP 2

Print the base

300 × 250 × 12 mm, bottom face on the bed. This is one of the two plates in this project whose first layer is a single unbroken circle metres long. Watch it.



Bottom face down, race and insert pockets up. This is the orientation as printed: nothing on it is rotated for the picture.

SETTINGS

Orientation	Bottom face on the bed
Walls	6
Top / bottom	6 layers
Infill	25% gyroid
Support	None
Brim	**Not inside the race**

WATCH THE FIRST TWO LAYERS

The 165 mm ball race round this plate means a point on that loop cools for a full lap before the nozzle comes back to it. Two plates failed there in the first layer or two while every other part in the tree came off the same profile clean. Stay with it until layer three is down.

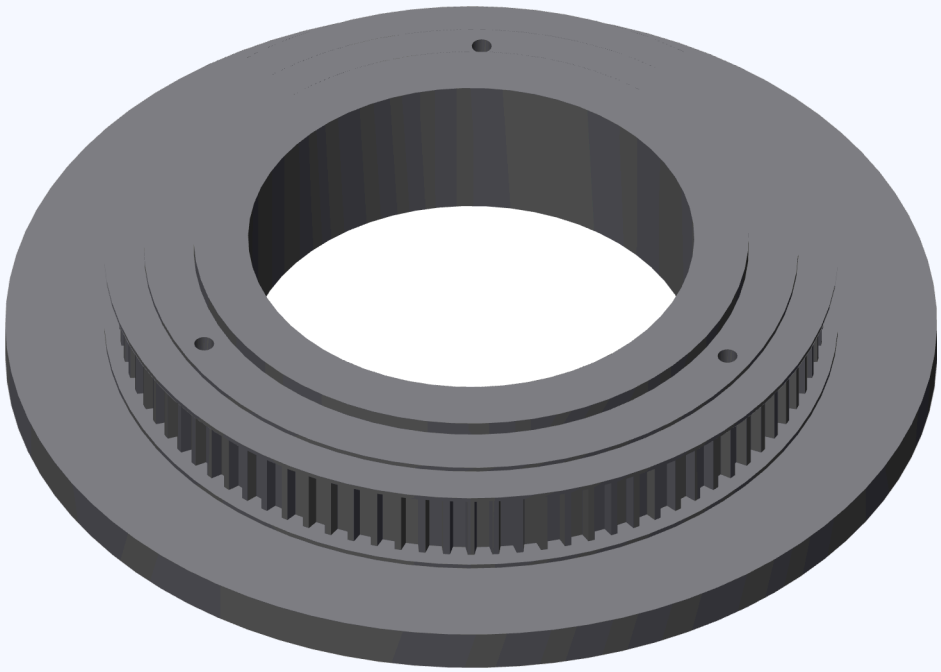
GATE — BEFORE IT COMES OFF THE PLATE

The race groove is one continuous, unlifted ring. Deburr it with a **plastic scraper only**. **Anything abrasive changes the groove's radius, and a changed radius becomes axial runout you cannot adjust out.**

PRINT · STEP 3

Print the turntable

The second long-loop plate. Flat underside on the bed, the 90-tooth pulley and the nest pedestal upward.



Underside down. That flat face is the datum the race ring and the spool both bolt to, so it is also the bed face.

SETTINGS

Orientation	Flat underside on the bed
Walls	6
Infill	30% gyroid
Support	**Disabled**

Nothing on this part needs support. Turn it off rather than leaving it on automatic: a support that lands in a pulley groove or an insert pocket is worse than anything it would have held up.

ALSO PRINT NOW

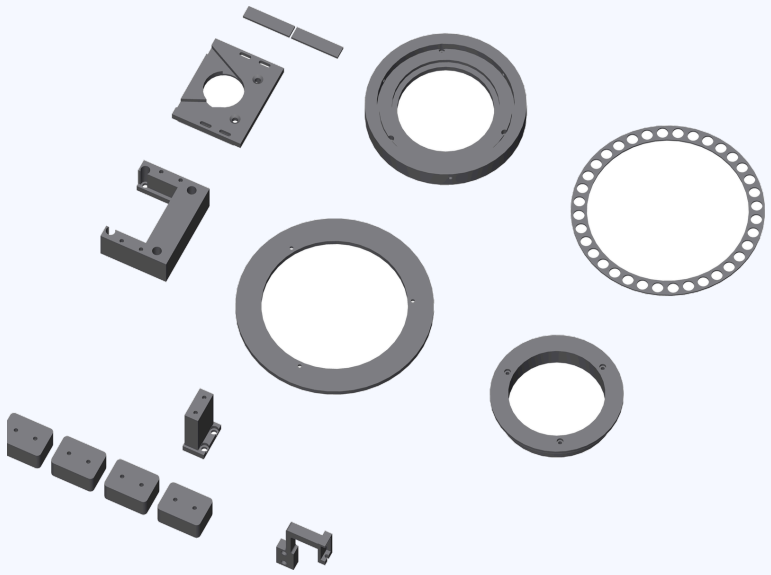
race-ring	Flat back on the bed, groove upward, **100% infill**

Deburr the ring groove with a plastic scraper only, for the same reason as the base.

PRINT · STEP 4

Print the rest

Eleven more shapes, each lying on the face it prints on. That face is the setting; the rest is ordinary.



Each part on the face that goes down.

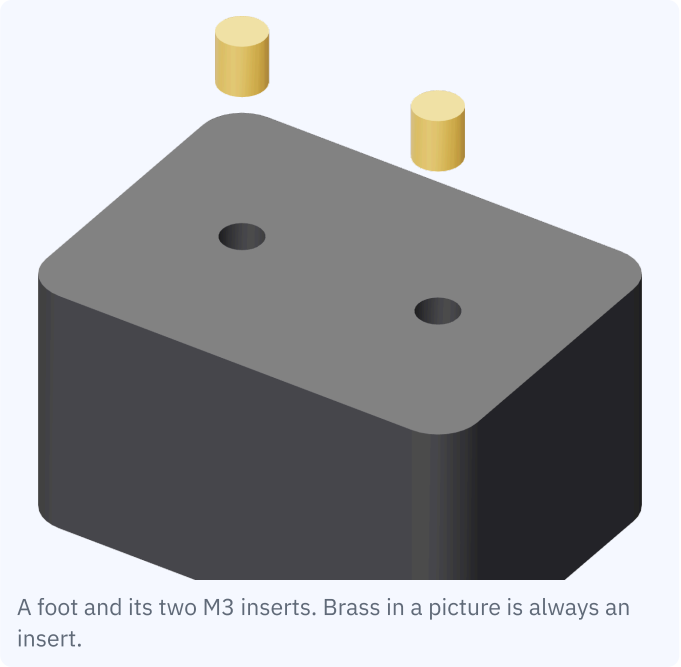
PART	ON THE BED	SETTING
base-foot × 4	Broad face	5 walls, 40% gyroid. Keep both screw bores clear.
spool	Flange	Hub upward, 5 walls, 30% gyroid.
motor-tower	On its feet	Rails upward. The four M5 access holes print as vertical bores.
motor-carriage	Broad arm underside	Skin and clamp walls up. **Build-plate-only tree support** under the two belt-clearance roofs.
motor-clamp-pad × 2	Either 48 × 12 mm face	Keep the screw-tip sockets clean.
ground-tower	Its foot	Ordinary.
ground-arm	Flat	Leaf and shoe fork in the bed plane. **100% infill.** Reject any layer separation.
tube-nest	Its register face	0.20 mm layer, 6 walls, **100% infill**. Keep the three radial passages clear.
ball-cage	Flat	Ordinary.

THE CARRIAGE'S TWO ROOFS

Supporting them keeps the Ø38.6 mm pilot ring continuous, and that ring — not the mounting screws — carries belt tension.

Thirty-seven inserts

Six M5 and thirty-one M3, all brass heat-set, all pressed before any part meets another part. Do them all in one sitting with the iron already hot.



A foot and its two M3 inserts. Brass in a picture is always an insert.

How each one goes in

1. Iron to temperature with the matching insert tip fitted.
2. Sit the insert on its bore, knurl down.
3. Bring the tip down straight and let the insert sink under the iron's own weight. Do not push.
4. Stop when the rim is level with the face. Lift the iron off and leave it alone for twenty seconds.

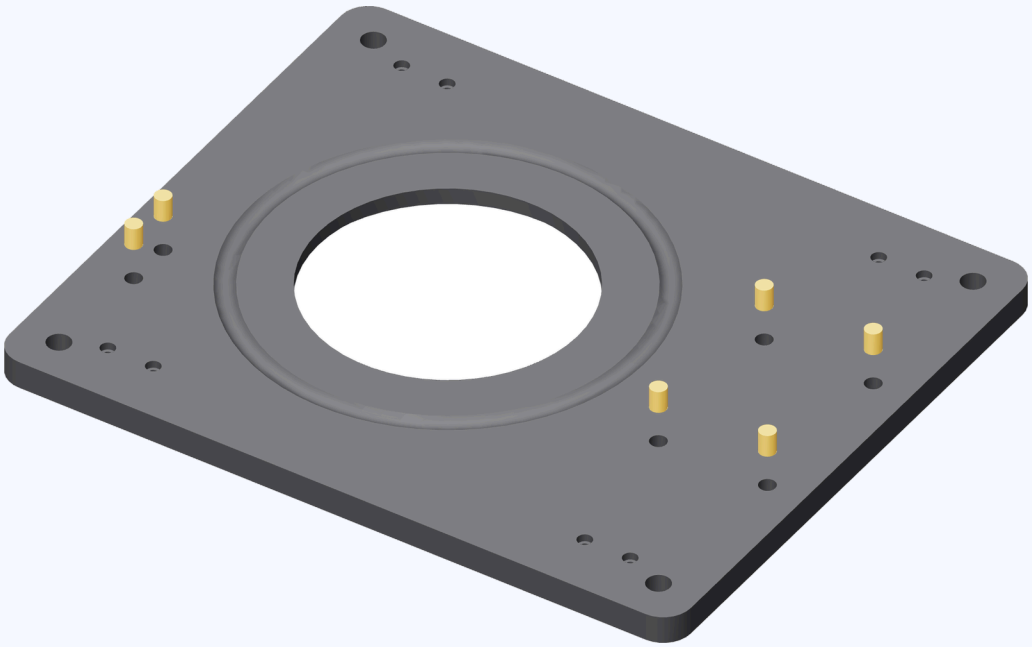
WHERE	COUNT	SIZE	DIRECTION
Base — tower and ground-tower stations	6	M5 × 9.5	Down into the top face
Turntable — underside	6	M3 short	Up into the flat underside
Turntable — nest register face	3	M3 short	Down
Nest — outer collar	3	M3 short	Inward, from the outside face
Motor tower — rail tops	4	M3 short	Down
Motor carriage — two outside walls	4	M3 short	Inward
Ground tower — top	2	M3 short	Down
Ground arm — clamp jaw outside face	1	M3 short	Inward
Base feet — two per foot	8	M3 short	Down into the top of each foot

TORQUE, LATER

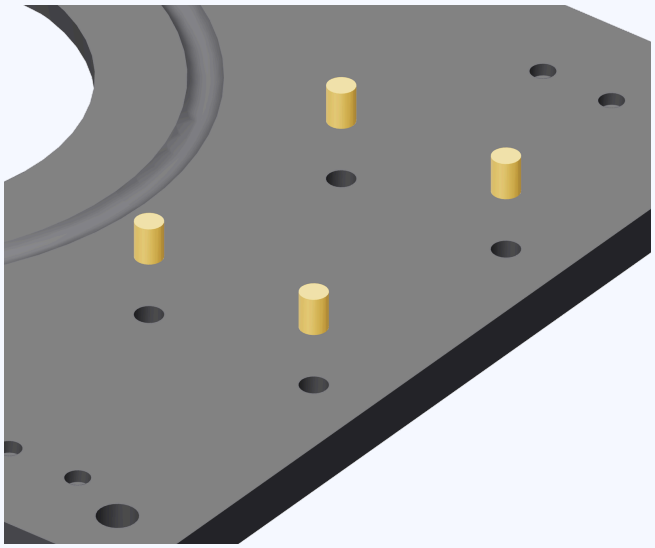
A ruthex insert's grip on this stock gives out well below what a 12.9 M3 or a drill's lowest clutch click will deliver. Every screw that lands in one of these inserts goes in on the **HOTO PixelDrive at its 0.5 N·m setting** or by hand. The torque the tool *will not* exceed is the spec that matters here.

The base: six M5

Four in a rectangle where the motor tower lands, two in a close pair at the far left where the ground tower does. All six go down into the base's top face.



The whole plate. Four tower stations at the right, the two ground-tower stations at the left edge. These are the only M5 inserts in the fixture.



The four tower stations, close.

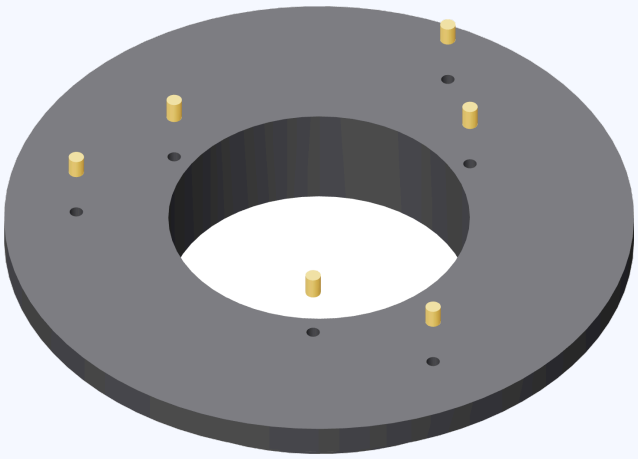
1. Fit the M5 tip. These bores are Ø7.0 mm and 9.7 mm deep — deeper than any M3 pocket on the fixture.
2. Press all four tower inserts first, then the two ground-tower inserts.
3. Sight across the plate. Every rim level with the top face.

NOT THE BENCH HOLES

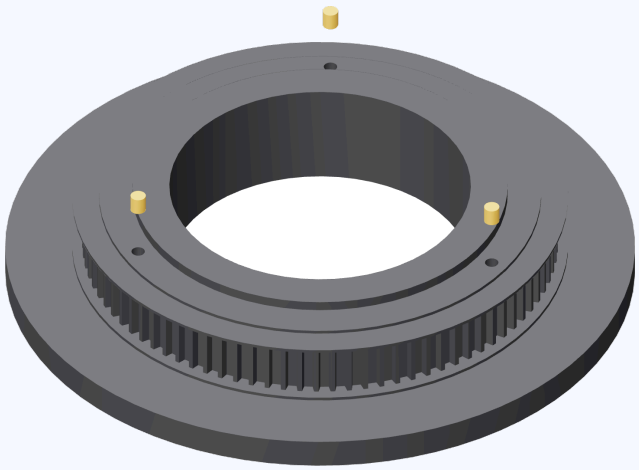
The four Ø10 mm holes near the corners are for clamping the fixture to the bench. Nothing goes in them.

The turntable: nine

Six up into the flat underside, three down into the nest register face on the other side. Turn the part over once, between the two sets.



Turned over. Three race-ring inserts on the outer circle, three spool inserts on the inner one.



Right way up. Three nest-retainer inserts in the raised register face.

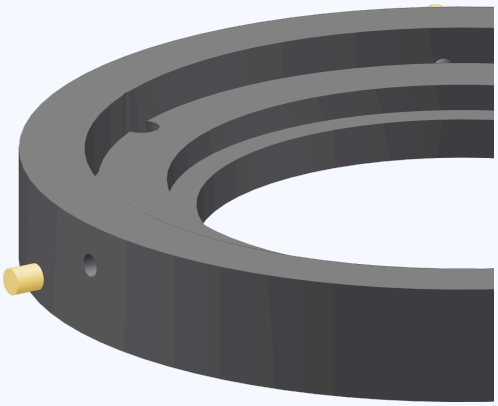
1. Turn the turntable **underside up** on a clean flat surface. That face is a print datum — nothing gritty under it.
2. Press three inserts on the **outer** circle, at 30°, 150° and 270°. These are for the race ring, and their pockets are 6.0 mm deep.
3. Press three on the **inner** circle, at 60°, 180° and 300°. These are for the spool, and their pockets are 6.8 mm deep.
4. Turn the turntable back over.
5. Press three into the raised nest register face, at 0°, 120° and 240°, 5.2 mm deep.

GATE — THE UNDERSIDE STAYS FLAT

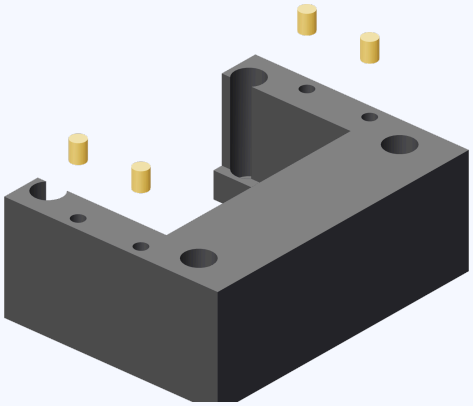
Lay a straight edge across the underside over each insert. **No rim may stand proud of that face.** The race ring bolts flat to it, and a high insert becomes a tilted race, which becomes runout at the weld circle.

The rest

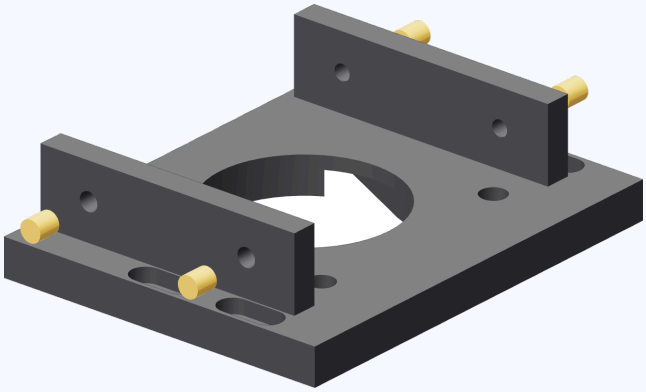
Nest, motor tower, motor carriage, ground tower, and the four feet. Four of these go in *sideways*, which is the only thing on this page that is not routine.



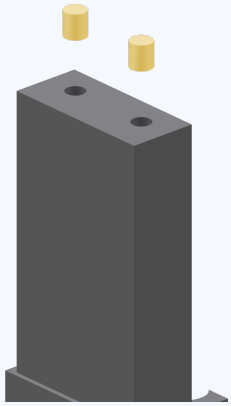
Nest — 3. Inward from the collar's outside face, 5.2 mm deep, at 60°, 180° and 300°. **Sideways.**



Motor tower — 4. Down into the two rail tops, 6.0 mm deep.



Carriage — 4. Inward from the two outside walls, 5.2 mm deep. **Sideways.**



Ground tower — 2. Down into its top face. The arm bolts to these.

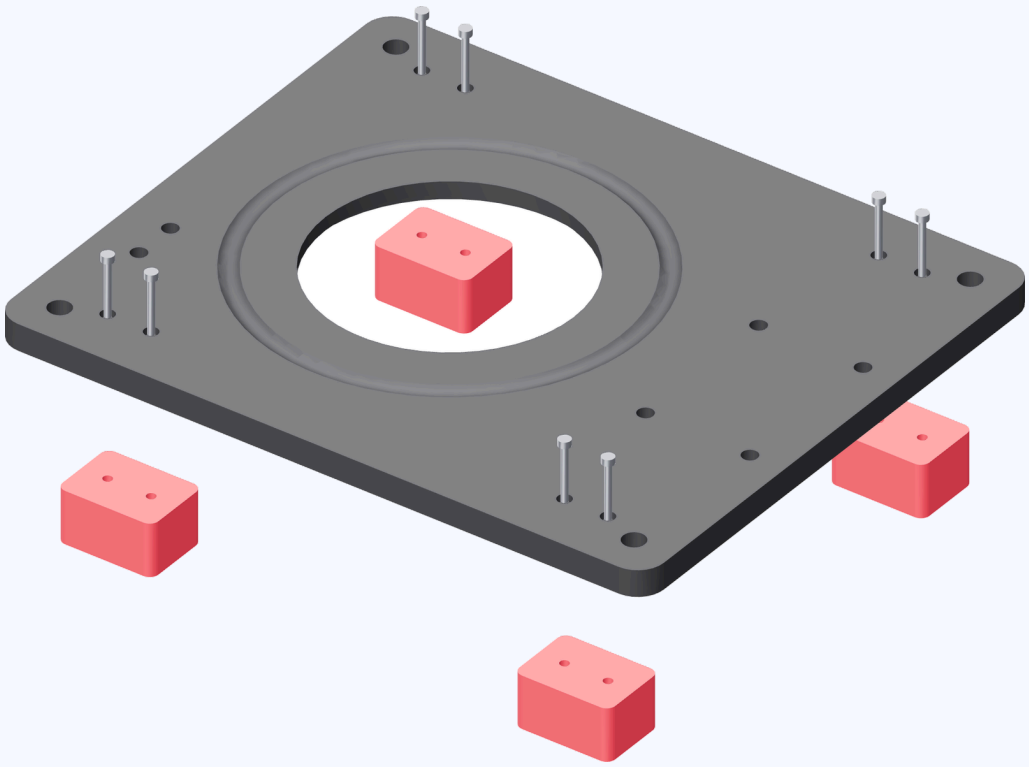
ALSO	COUNT	WHERE
Ground arm	1	Inward, through the clamp jaw's outside face. This is the one that grips the copper. **Sideways.**
Base feet	8	Two down into the top of each foot, 5.2 mm deep.

A SIDEWAYS INSERT WANTS A REST

Lay the part on the bench so the bore points *up* and press vertically. Trying to drive one horizontally with the part standing is how a knurl ends up crooked, and a crooked insert in the nest collar puts an adjuster screw off the tube's radius.

Put the base on its feet

Eight M3 × 25, driven down through the base's top face into the feet below it. The base is never turned over for this.



The four feet stand on the bench; the base comes down onto them; the eight screws go in from above. Coral is what this step adds.

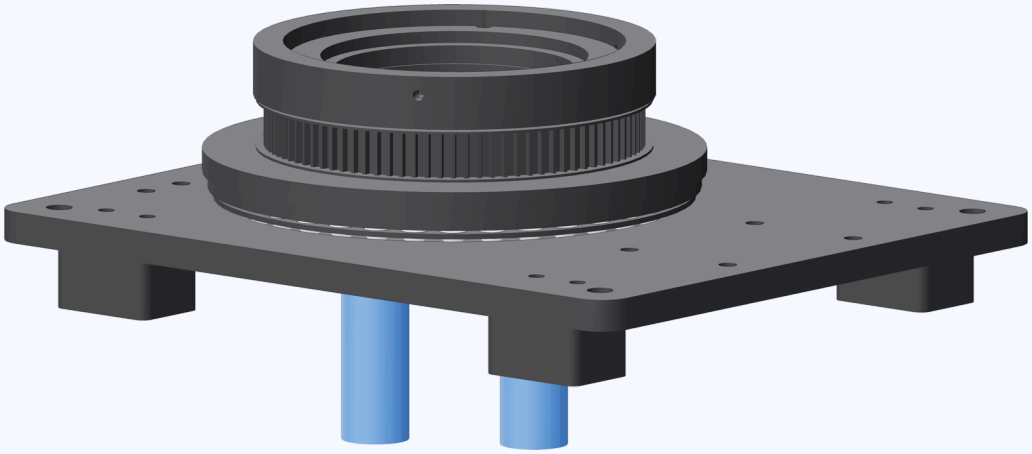
1. Stand the four feet on the bench roughly where they go — two at each end, their long side across the base.
2. Lower the base onto them and shuffle each foot until both of its screw bores line up with the counterbores in the base.
3. Drop all eight M3 × 25 in first and start every one by a turn or two before tightening any.
4. Bring them down on the PixelDrive at **0.5 N·m**, working diagonally across the plate. Each head sits flush in its counterbore.
5. Stand the fixture on the bench and rock it with a hand on each end.

GATE — IT SITS FLAT

All four feet touch the bench and the base does not rock. **Nothing projects below any foot.** The base now stands 24 mm off the bench, and that 24 mm is what makes the second closure's purge connection possible.

Prove the service path

A Ø90 mm passage runs all the way through the nest, turntable, spool and base. Check now that a 25.4 mm envelope clears at both port positions, while it is still a plate and four feet.



Two 25.4 mm probes standing where the already-welded lower plate's two recessed ports will sit. They must pass the base and the spool without touching either.

1. Take a 25.4 mm round bar, a deep socket, or anything of that diameter and about 150 mm long.
2. Pass it up through the base's service opening from below, at a radius of about 30 mm from the axis.
3. Walk it round to the other port position, 180° away, and pass it again.
4. Look under the fixture. There must be clear air from the bench to the passage at both positions.

GATE — 25.4 MM, TWICE

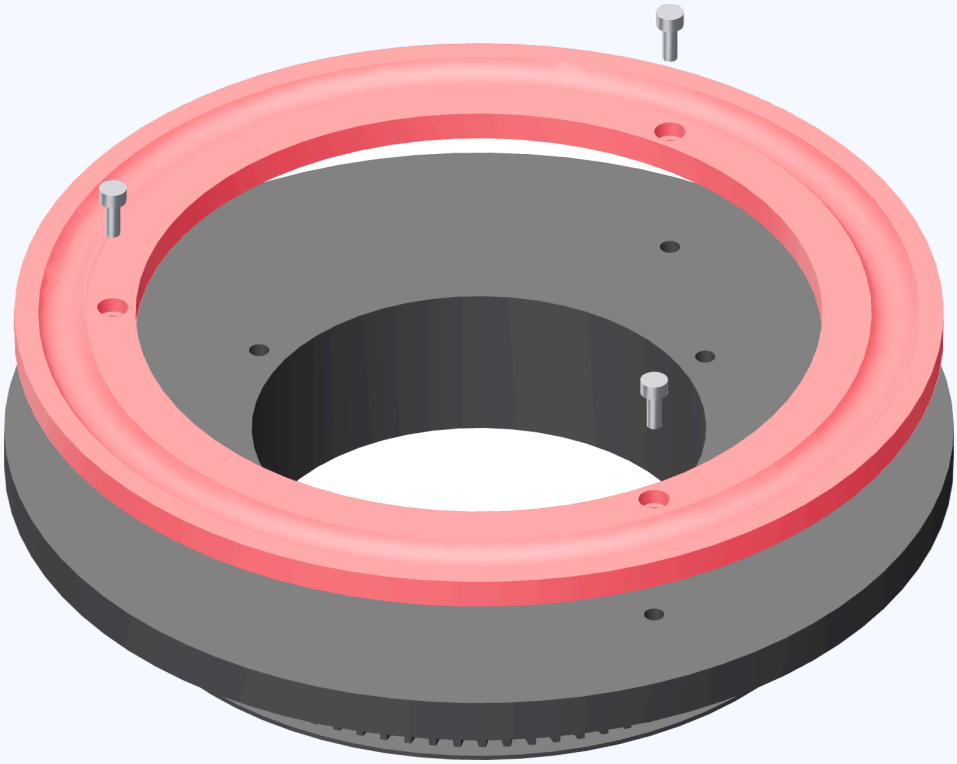
Neither the base nor the spool obstructs the probe at either port position. **If it fouls, find out now.** When the second closure is on the table there is a welded vessel sitting on the nest and a purge hose already connected to one lower port, and this is not a thing you can discover then.

WHY IT EXISTS

The first closure back-purges through the open held end. The second cannot — that end is welded shut. So the purge goes in through one port of the completed lower plate and out through the other, both of them reached through this passage from underneath.

The race ring, groove down

The ring printed groove-up, because a ball's running surface should be a first-layer flat. It bolts on groove-*down*.



The turntable turned over. The ring comes down onto the flat underside with its groove facing away from the part — toward where the balls will be.

1. Turn the turntable underside up.
2. Sit the race ring on it with the **grooved** face upward in this position — which means downward, toward the balls, when the fixture is the right way up.
3. Line the three screw holes up with the three outer-circle inserts at 30°, 150° and 270°.
4. Drive three flush M3 × 8 by hand or at 0.5 N·m.
5. Run a fingertip round the groove over each screw.

GATE — NOTHING PROUD IN THE GROOVE

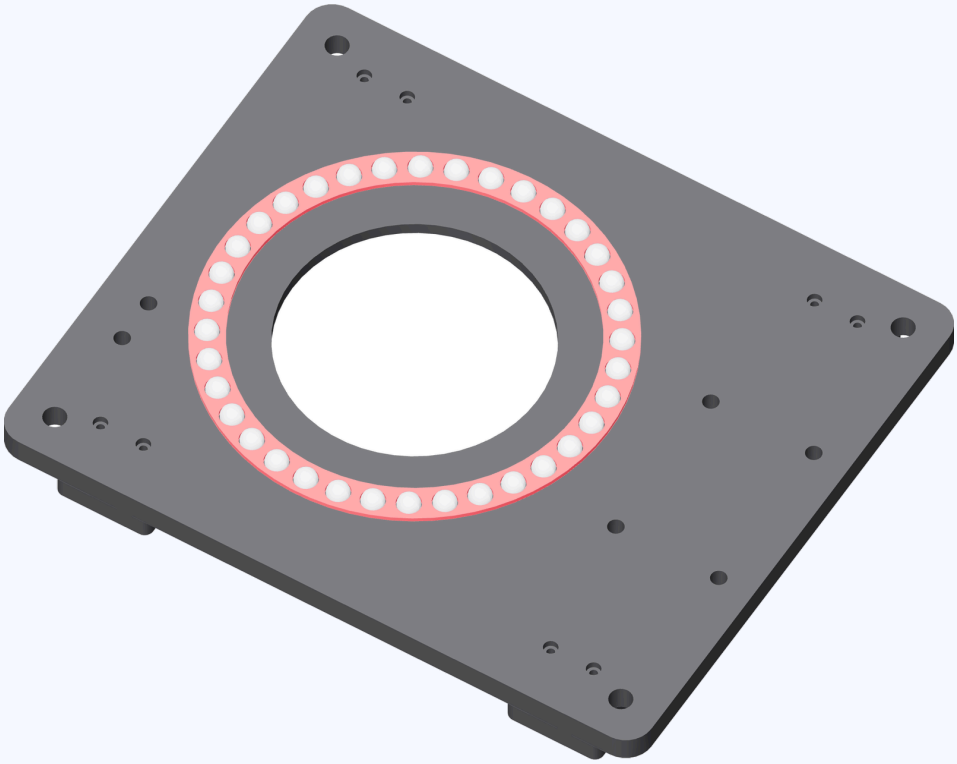
No screw head stands above the ring's groove face. **A proud head is a ball riding over a bump once per revolution**, and that shows up as a periodic wobble in the indicator later that no adjuster screw will remove.

PLASTIC ONLY

If the groove needs deburring, a plastic scraper. Abrasive changes the groove radius, and the groove radius is the bearing.

Sort and load the balls

Thirty-six 10 mm polypropylene balls out of the bag of two hundred. You are choosing thirty-six that agree with each other, not thirty-six at random.



The cage laid in the base's lower race with all thirty-six pockets filled. The cage spaces the balls 14.4 mm apart on a 165 mm circle and carries no load.

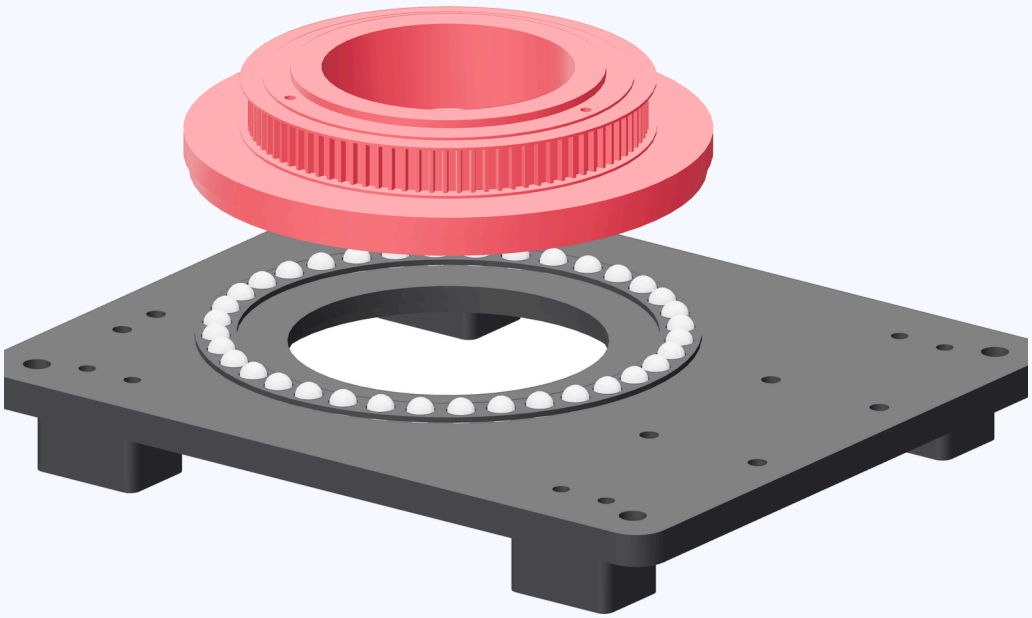
1. Measure twenty or so balls with the calliper and note the median diameter.
2. Set the fixture flat and level on the bench.
3. Lay the cage into the base's lower race groove.
4. Fill all thirty-six pockets, measuring as you go. **Reject any ball more than 0.05 mm off the median**, or any that differs visibly in colour, gloss or shape. You have 164 spares.
5. Roll each seated ball with a fingertip. It should turn freely in its pocket.

GATE — THIRTY-SIX THAT AGREE

Every pocket filled, every ball within 0.05 mm of the median. **One oversize ball carries the whole vessel by itself** and lifts the platter off the other thirty-five once per revolution.

The turntable down

Straight down onto the loaded race, level, with the ring's groove finding the balls all the way round at once.



The turntable and its race ring as one piece, coming down onto thirty-six balls. Keep it level: a tipped platter drops one edge into the groove and pushes balls out of the cage at the other.

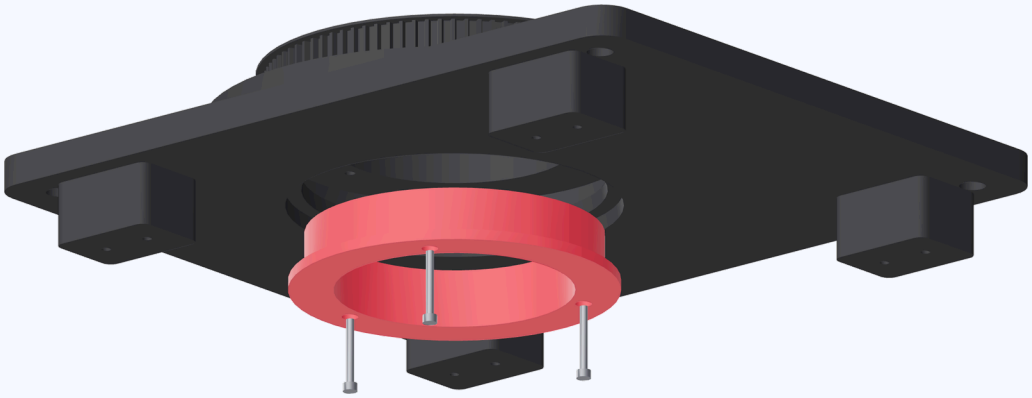
1. Pick the turntable up by its rim with both hands, ring downward.
2. Bring it over the base and align the ring's circle with the ball circle by eye.
3. Lower it **level** and let it settle on its own weight. Do not press.
4. Look round the joint. The ring's groove should sit on the balls with an even gap to the base all the way round.
5. Turn the platter slowly by hand, one complete revolution, then back.

GATE — IT TURNS FREE AND QUIET

One revolution each way with a fingertip's push and no binding, no grinding, no catching at any one angle. **A regular catch once per revolution is a ball, a proud screw, or a burr in a groove** — lift the platter and find it.

The spool, up from below

The spool does not carry the running load. It catches lift, and it keeps the Ø90 mm passage continuous through the base.



Seen from underneath. The hub passes up through the base's Ø114 mm bore and three M3 × 25 draw it into the platter above.

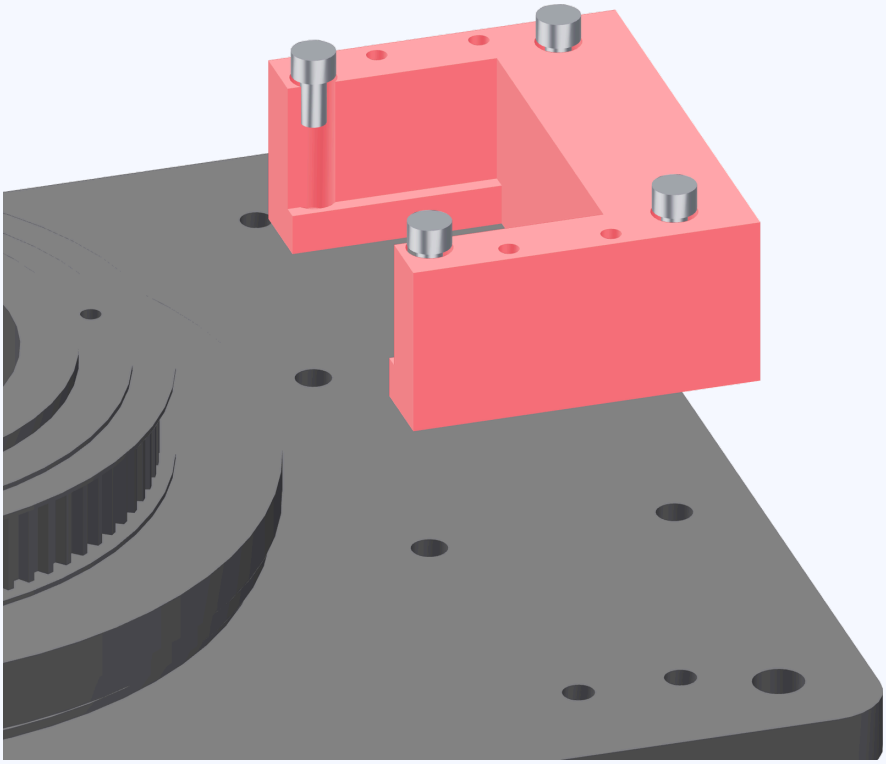
1. Turn the fixture on its side, or get under it — the feet leave 24 mm, which is enough for a driver but not for a hand and a driver.
2. Pass the spool hub up through the base's central bore from below, flange down.
3. Turn it until its three flange holes find the platter's three inner-circle inserts at 60°, 180° and 300°. Turning the *platter* is easier than turning the spool.
4. Start all three M3 × 25 before tightening any, then bring them up evenly at 0.5 N·m.
5. Set the fixture back on its feet and look at the flange rim.

GATE — THE FLANGE FLOATS

The flange rim sits inside the base's recess, about **1 mm clear** of the base shoulder, and **nothing projects below the base**. Turn the platter a revolution: it must still be free. **If the spool has taken up the running load, the balls are now unloaded and the axis is a flexible hub** — back the three screws off and re-seat.

Bolt down the motor tower

Two rails and a rear wall, on the base's four M5 stations. It goes on now; the carriage stays off it until the motor is bolted to the carriage.



Two rails, and the rear wall behind them. Everything about this part is shaped to stay outside the belt's swept path across the carriage's whole 8 mm of travel.

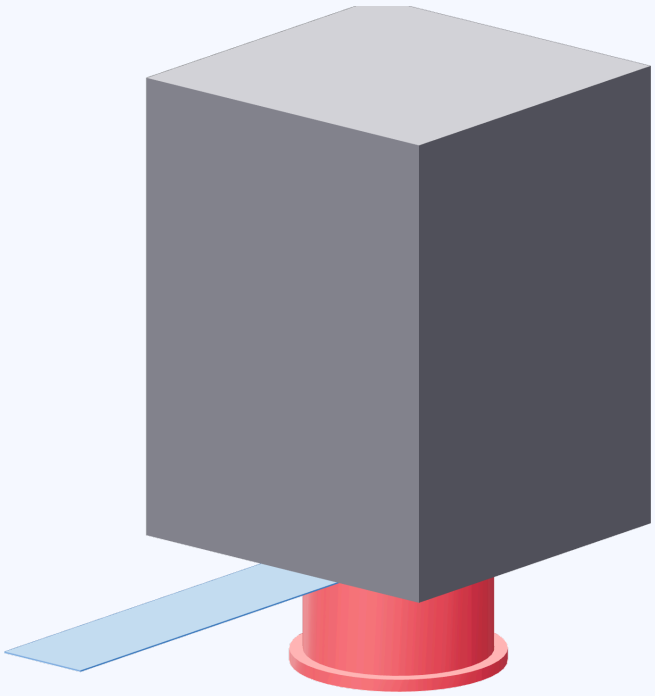
1. Set the tower on its four stations, rails upward, rear wall away from the tube.
2. Drop four M5 × 10 socket-head screws down the tower's own access holes and start each in its insert.
3. Tighten with the 4 mm hex down the access holes, opposite corners first. **These land in brass inserts** — hand-tight and no more.
4. Push the tower sideways with a hand. It should not move at all.

LEAVE THE CARRIAGE OFF

The next two steps happen at the bench, off the fixture: the pulley goes on the shaft, then the motor bolts to the carriage. Once the carriage's arms are on these rails, the tower's rear wall stands under the motor's two countersinks and no driver reaches them.

The pulley on the shaft

This is the one dimension on the fixture you set with a feeler blade, and it is what puts the belt in the middle of the printed tooth zone.



A 0.25 mm blade between the pulley's motor-side flange and the front face of the motor's 1.6 mm Ø38.1 mm pilot.

The datum is the pilot

Not the shaft end. The pulley's outer face stands **0.85 mm beyond** the nominal 21 mm shaft end, so a pulley pushed flush to the shaft tip is a pulley in the wrong place.

Gap to set	0.25 mm
Pilot diameter	Ø38.1 mm
Pulley bore	6.35 mm
Set screws	2, radial

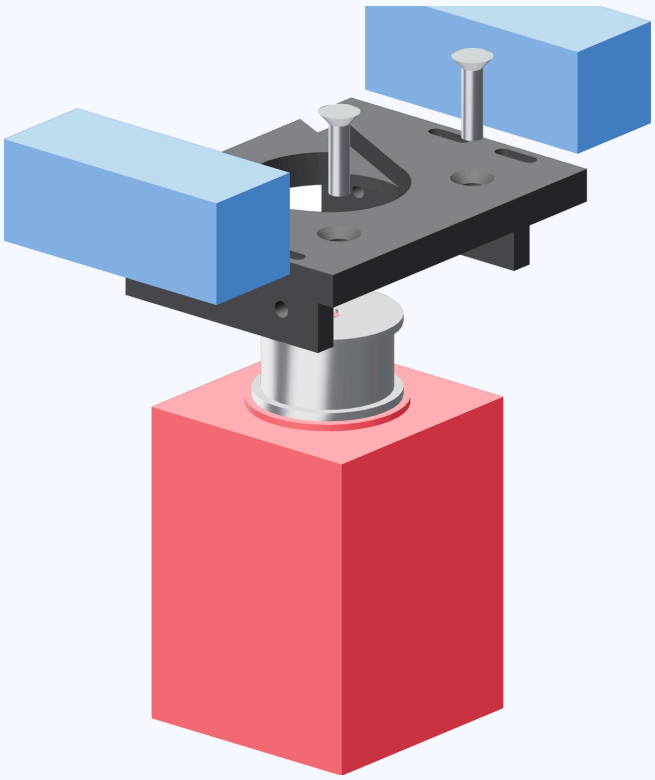
1. Measure the pulley's **actual** overall length with the calliper. Nominal is 20 mm; use what you measure.
2. Slide it onto the shaft, flanged end toward the motor, until the distance from the pilot's front face to the pulley's outer face is **that measured length plus 0.25 mm**.
3. Or, equivalently: slip a 0.25 mm / 0.010 inch feeler blade between the motor-side flange and the pilot and close the pulley onto it.
4. Turn the pulley until one set screw is over the shaft's **D-flat**. Tighten that one first.
5. Tighten the second set screw against the round part of the shaft.
6. Withdraw the blade and spin the pulley by hand.

GATE — AN AIR GAP THAT STAYS

The pulley turns with the shaft, and there is visible daylight between its motor-side flange and the stationary pilot. **If the flange rubs the pilot, the belt is 0.25 mm low and the pulley is dragging on the motor.**

The motor on the carriage

On the bench, before either piece meets the tower. The motor hangs face-down and the carriage's 2 mm skin takes its face pilot.



Carriage skin-up on two 25 mm blocks, so the pulley has the 13.85 mm it hangs below the arms. Two M5 × 20 rise through the arms from underneath.

1. Stand the carriage skin-up on two blocks about 25 mm tall, arms bridging the gap between them.
2. Lower the motor face-down. The pulley passes through the skin's Ø38.6 mm hole and the Ø38.1 mm pilot seats in it.
3. Turn the motor until its **rear** pair of flange holes lines up with the two countersinks in the arms. The front pair stays empty — the belt's swept path crosses it at full tension.
4. Pass an M5 × 20 countersunk screw up through each arm and through the motor's Ø5.2 mm flange hole.
5. Drop an M5 square nut into the motor's open corner channel above each screw and engage it. The nut's flats sit on the channel's two walls and hold it.
6. Pull both down evenly with the **3 mm** hex. Seat them by the heads: each 9.8 mm head must sit flush inside its recess in the arm's underside.

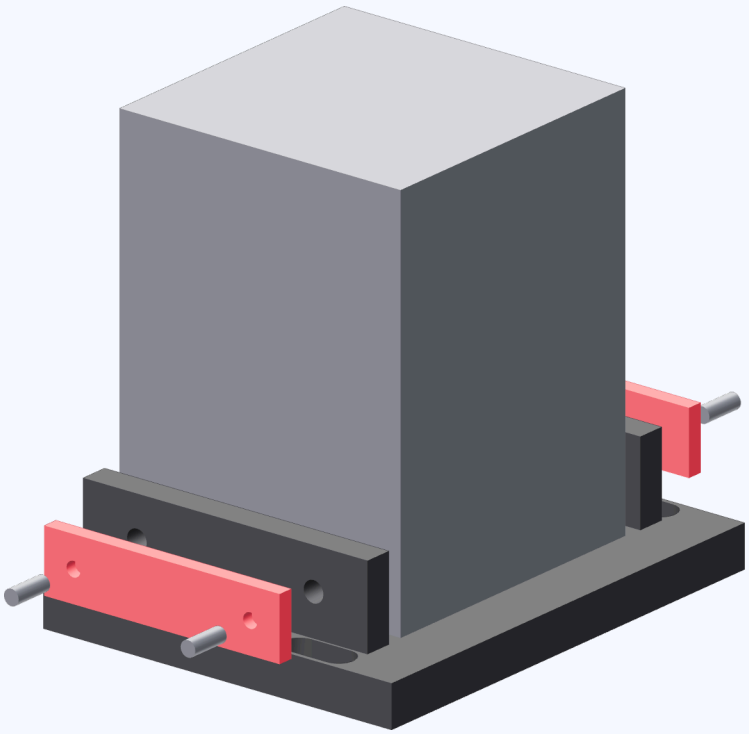
THE SCREW DOES NOT THREAD INTO THE MOTOR

The 23HS30-2804S's four flange holes are **Ø5.2 mm clearance**, not tapped. An M5 × 12 reaches past the 8 mm arm and ends 1 mm below the top of the 5 mm flange ear, where nothing can catch it. It needs the 20 mm screw and the square nut: 8 mm arm, 5 mm ear, 4 mm nut, 3 mm proud. **Confirm the nut seats in the channel on the first assembly** — that fit is recorded from the catalogue, not yet from this bench.

DRIVE · STEP 14

The two clamp pads

They stop the motor rocking. They do not hold the belt tension — the pilot ring in the skin does that.



One pad down each Y side of the 57.3 mm frame, each taking two M3 × 8 inward through the carriage's side walls.

1. Slide one pad down each side of the motor frame, between the frame and the carriage's side wall.
2. Start all four M3 × 8 screws inward through the walls' inserts, into the pads' screw-tip sockets.
3. Advance them **evenly**, a turn at a time, alternating sides and ends.
4. Stop as soon as the motor cannot rock. Try to twist the motor in the carriage: no movement, and no bowed side wall.

EVEN, AND LIGHT

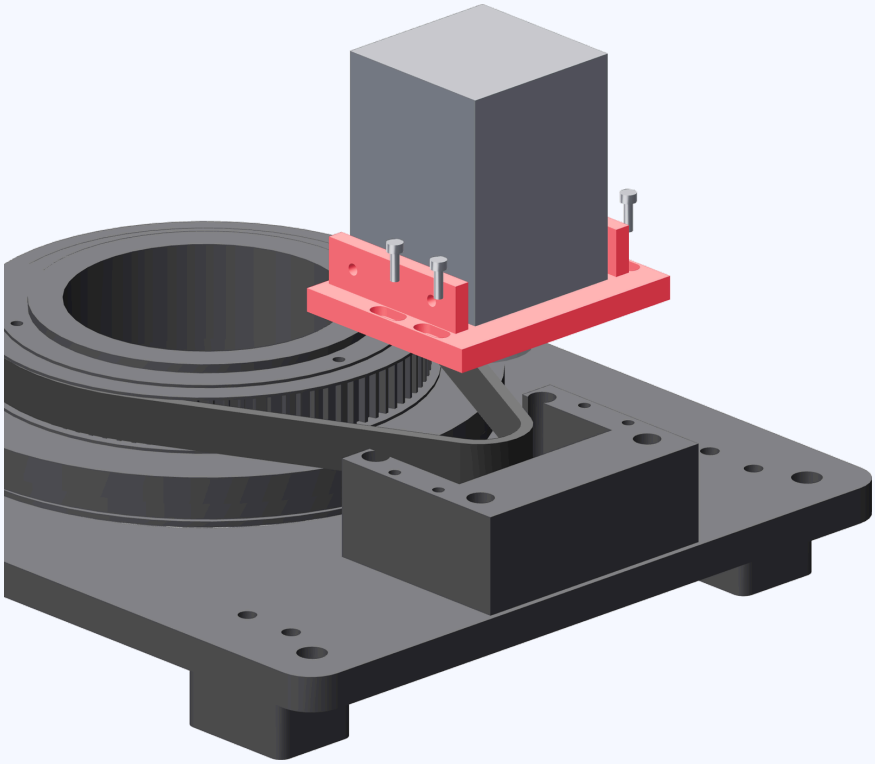
These four screws land in brass inserts in a 3 mm printed wall. Over-tightening one side pushes the motor off the pilot it is registered on, which is the one thing this joint must not do.

GATE — THE ASSEMBLY IS ONE PIECE

Pick the carriage up by the motor. Nothing shifts, nothing rattles, and the pulley still spins free of the pilot. The carriage and motor now go onto the tower together.

Into the belt, onto the rails

The belt goes on the big pulley first and is held open. The carriage and motor come down as one piece, and the small pulley enters the loop on the way.



Four M3 × 10 through the carriage's slots into the rail-top inserts. Start them at the **inboard** end of the slots and leave them loose.

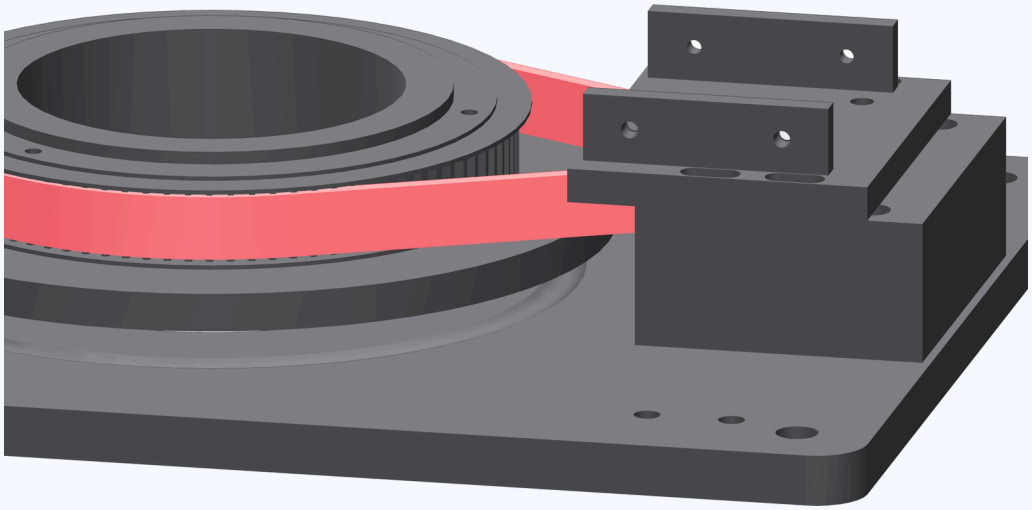
1. Hang the 550-5M-15 belt on the turntable's 90-tooth pulley, teeth inward, and let it find its grooves all the way round.
2. Hold the free bight open with one hand, wide enough for the 20-tooth pulley to pass into it.
3. Bring the carriage and motor over as one piece, small pulley pointing into the bight.
4. Lower until the 20-tooth pulley is inside the loop and the carriage's arms land flat on the two rail tops.
5. Start four M3 × 10 through the carriage's slots into the rail inserts, at the **inboard** end of each slot. **Leave them loose.**
6. Check the belt is in the printed grooves for its whole wrap and between both flanges of the small pulley.

DO NOT LEVER THE BELT ON

Never roll the belt onto the small pulley with a screwdriver against the flange. The slots exist so the belt goes on slack and the carriage moves outward afterwards.

Belt tension

Slide the whole carriage outward until the long span twists about ninety degrees under finger pressure. That is the specification; there is no gauge.



The two spans. The **long** one — the outer of the two — is the one you twist. The slots give 8 mm of outward travel from 1 mm inside the nominal 125.1 mm centre distance.

1. With the four M3 × 10 still loose, pull the carriage outward, away from the turntable, with steady even pressure on both arms.
2. Pinch the middle of the long span between finger and thumb and twist it.
3. Adjust until it twists about **90°** and no further. Slack enough to twist to 180° is too loose; stiff enough to refuse 45° is too tight.
4. Holding that position, tighten the four M3 × 10 — diagonal pairs, at 0.5 N·m.
5. Re-check the twist. Tightening the screws can pull the carriage back.

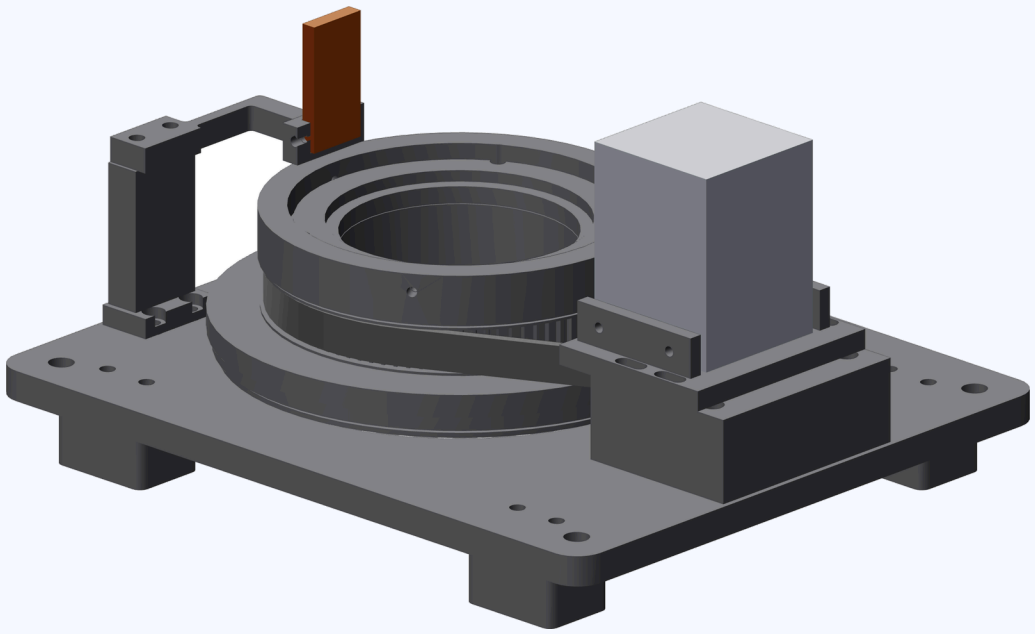
GATE — TRACKING

Turn the turntable by hand through two full revolutions. The belt stays **between both flanges** of the 20-tooth pulley and inside the printed tooth zone of the 90. **A belt that walks toward one flange is a carriage that is not square on its rails** — loosen all four, re-seat the arms flat, and tension again.

DRIVE · STEP 17

Ten revolutions, by hand

The last thing before the nest goes on. Nothing here needs power. You are running the bearing in and looking for the one fault that is cheap now and expensive later.



What you should have: base on feet, bearing loaded, spool up, tower bolted, motor and belt tensioned. No nest, no ground tower, no tube.

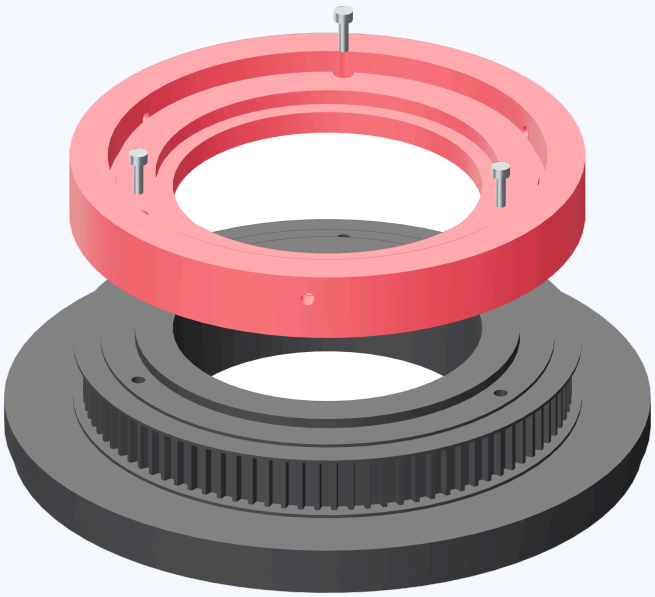
1. Turn the turntable slowly through **ten full revolutions clockwise**, watching the belt at the small pulley.
2. Turn it **ten more counterclockwise**.
3. Lift the platter off, wipe the lower race and the cage with a dry cloth, and look at what came out. A little PET-GF dust is expected on a first run-in.
4. Put it back on, level, and turn it another two revolutions.
5. Look at the spool flange from underneath as it turns.

GATE — THREE READINGS AT ONCE

The belt stays between both flanges for all twenty revolutions. The race does not bind at any angle. The spool flange **stays unloaded** — it should never touch the base shoulder while turning. **Any one of the three failing stops the build here.**

The nest

The only tube-size-specific part in the fixture. A later tube size costs this one 150 mm print and nothing else.



Down onto the Ø112 mm annular register. Three M3 × 10 go straight down through access wells in the ID pilot.

Three surfaces

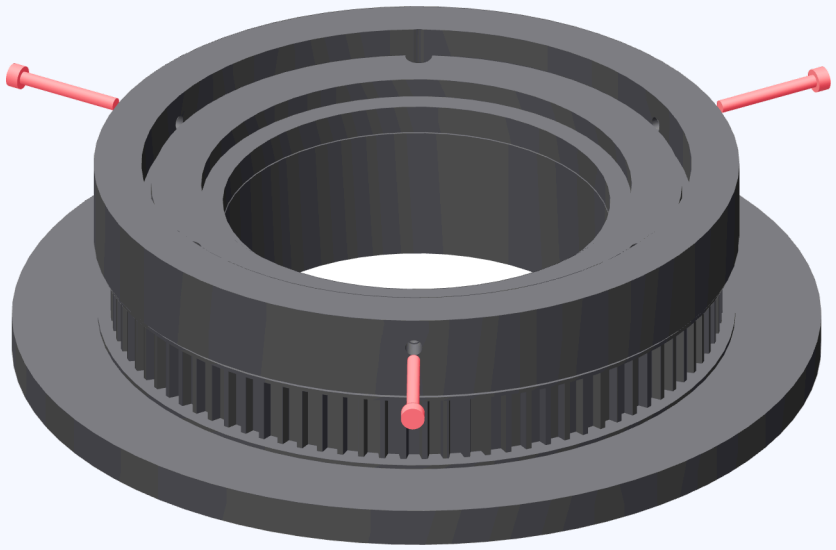
ID pilot	Ø123.30 × 4.5 mm — 0.20 mm radial clearance
OD guide	Ø127.80 mm — 0.40 mm radial clearance
Adjusters	3 × M3, at 120°, tips on the tube OD

The ID pilot is shorter than the end cap's 6.35 mm recess, so the same nest takes a bare tube end *or* an already-welded first end.

1. Sit the nest on the turntable's register and turn it until the three access wells line up with the three inserts at 0°, 120° and 240°.
2. Drop three M3 × 10 straight down the wells. **Use the slim-shank bit or an L-key** — a 1/4 inch hex bit body will not enter a Ø6.2 mm well.
3. Bring all three down evenly by hand. They engage the inserts by 3.2 mm; do not chase more.
4. Check each head is seated in its recess in the nest base.

The three adjusters

Thread them in now and then back every tip fully clear, so the first tube lands on the pilot and the guide rather than on three screw points.



Three M3 × 25 through radial inserts in the outer collar, at 60°, 180° and 300°. Their stations are staggered between the retainer wells, and the ID pilot backs the tube at every contact.

1. Thread one M3 × 25 into each of the three radial inserts from the collar's outside face.
2. Run each one **in** until you can see the tip appear in the guide bore, so you know the passage is clear and the screw is not cross-threaded.
3. Back every tip **fully clear** of the OD guide.
4. Leave them there. They are set against a tube, on page 43, and not before.

THEY LOCATE; THEY DO NOT CLAMP

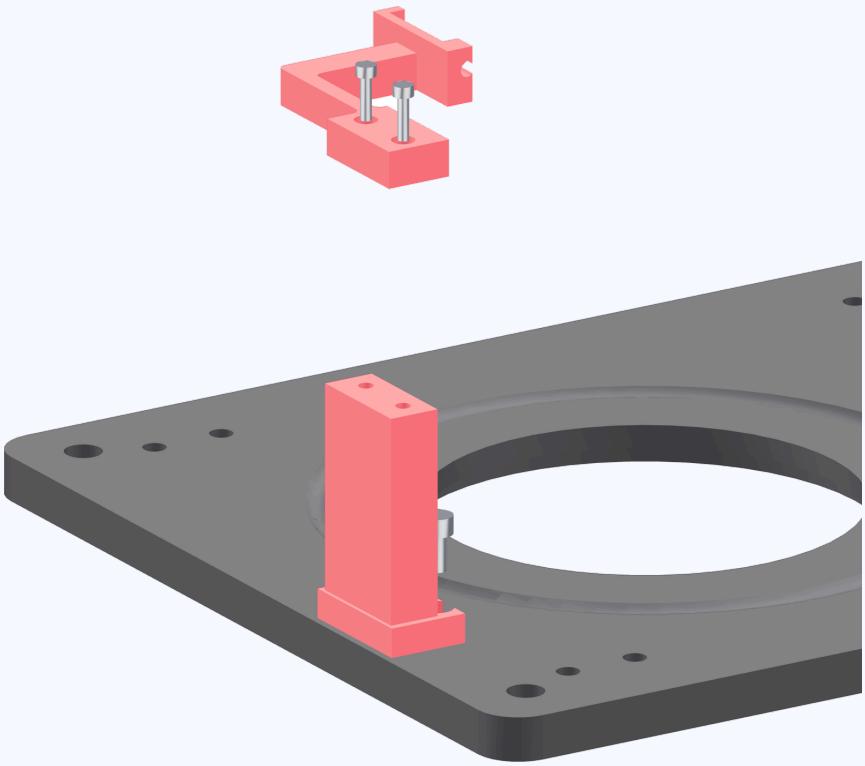
The tube wall is 0.065 inch. These three screws take up clearance and give the indicator something to adjust. **Light contact only.** They are not there to force a slightly oval tube round, and force applied here becomes shape you cannot weld to.

BRING THE SCREW TO YOU

The adjusters ride on the turntable. When one is on the far side, use the pedal to bring it round to an open side of the fixture rather than reaching across the tube.

The ground tower and arm

A stationary post at the far left of the base, and a replaceable leaf spring on top of it. This is what keeps the welder's work lead still while the vessel turns.



The tower on the base's two remaining M5 stations, and the arm coming down on two M3 × 12. Its rail feet seat the M5 heads at the same depth the motor tower's do.

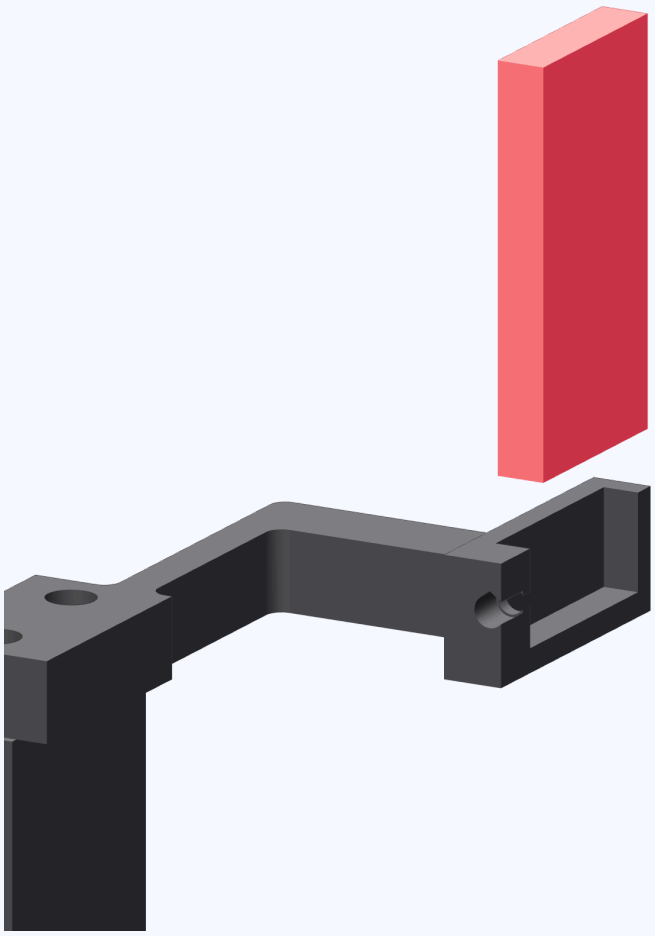
1. Set the ground tower on the two M5 stations at the left edge of the base, its fork end pointing at the tube axis.
2. Drive two M5 × 10 socket-head screws with the 4 mm hex. Hand-tight into brass.
3. Lay the flexure arm on the tower's top face, leaf pointing toward the tube axis, and line its two holes up with the tower's two inserts.
4. Drive two M3 × 12 at 0.5 N·m.
5. Push the leaf sideways with a finger. It should deflect and spring straight back with no creak and no visible layer line opening.

GATE — THE LEAF IS SOUND

It bends in plane and returns. **A whitening line, a crack, or a crunch when it flexes means the print delaminated** — this part is 100% infill for exactly that reason, and a failed one is a five-minute reprint.

The copper shoe

One 25 mm crosscut from the acquired 1/4 × 2 inch C110 bar. No hole is drilled in it and no thread is cut in it.



Down into the fork until it seats on the printed shelf. One factory broad face toward the tube; the saw cut is only a seated side.

Cutting it

1. Put the bar's **1/4 inch edge** on the BA4555 table, so the 2-inch face stands vertically.
2. Set the saw to its **125 FPM** low speed. The installed 24 TPI M42 blade then has about six teeth in the 6 mm cut width.
3. Cut one 25 mm slice with a light downfeed. Let the gullets clear; do not crowd it.
4. Break the cut edge's burr. Leave both broad faces alone — the factory surface is the tube contact.

The result is a 6 × 25 × 50 mm block, stood with the 50 mm mill width vertical.

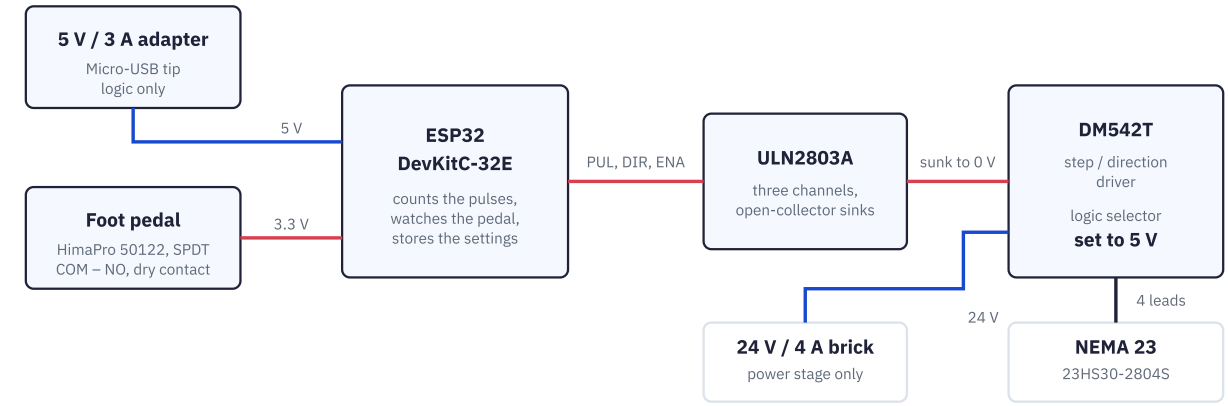
1. Slide the shoe down into the arm's fork until it seats on the printed shelf, one factory broad face toward the tube.
2. Tighten the single M3 × 8 side clamp only until the shoe cannot move in the fork.

THE PRELOAD COMES FROM THE TUBE

A loaded tube pushes the shoe back against the fork's rear wall and deflects the long leaf outward. The delivered bar's thickness window leaves 0.75–1.50 mm of preload, which is what keeps the contact live through runout. The clamp screw's only job is to stop the shoe sliding.

What the controller is

Two supplies that never meet, one dry-contact pedal, and an ESP32 that counts pulses. Read this page before you strip a single wire.



THE TWO SUPPLIES NEVER MEET

The 5 V adapter powers the ESP32 and all three 5 V signal loops. The 24 V brick powers *only* the DM542T's power stage and the motor. They stay separate across the driver's own opto-isolators. **Never connect 24 V to the ESP32.**

THE PEDAL IS NOT A POWER SWITCH

It carries 3.3 V through a 3.3 kΩ pull-up and nothing else. Opening the contact — releasing it, or pulling a lead out — always stops new pulses. The controller also refuses to move at boot until it has seen the pedal released once.

THE MOTOR IS HELD ONLY WHILE IT HAS WORK

The ESP32 works the DM542T's ENA input through the third ULN channel. The driver holds the motor while the table turns and for ten seconds after a release. Then it lets go: the motor is cold and the table turns by hand until the next press, which energises the coils before the first pulse.

Set the driver switches

Eight DIP switches on the DM542T. Set them with the driver unpowered, before anything is wired to it.



Coral is the lever pushed to the ON end of its travel. Check the silkscreen on your own driver for which end that is — it is marked.

What those eight mean

SW1–SW4	Peak current, **3.76 A**
SW5	Standstill current, **50%**
SW6–SW8	Microstepping, **3,200 pulses/rev**

3.76 A is the DM542T table's closest setting to the manual's *phase current × 1.4* peak rule for a 2.8 A/phase motor. The standstill reduction limits motor heating while you are setting the welder up.

Also on the driver

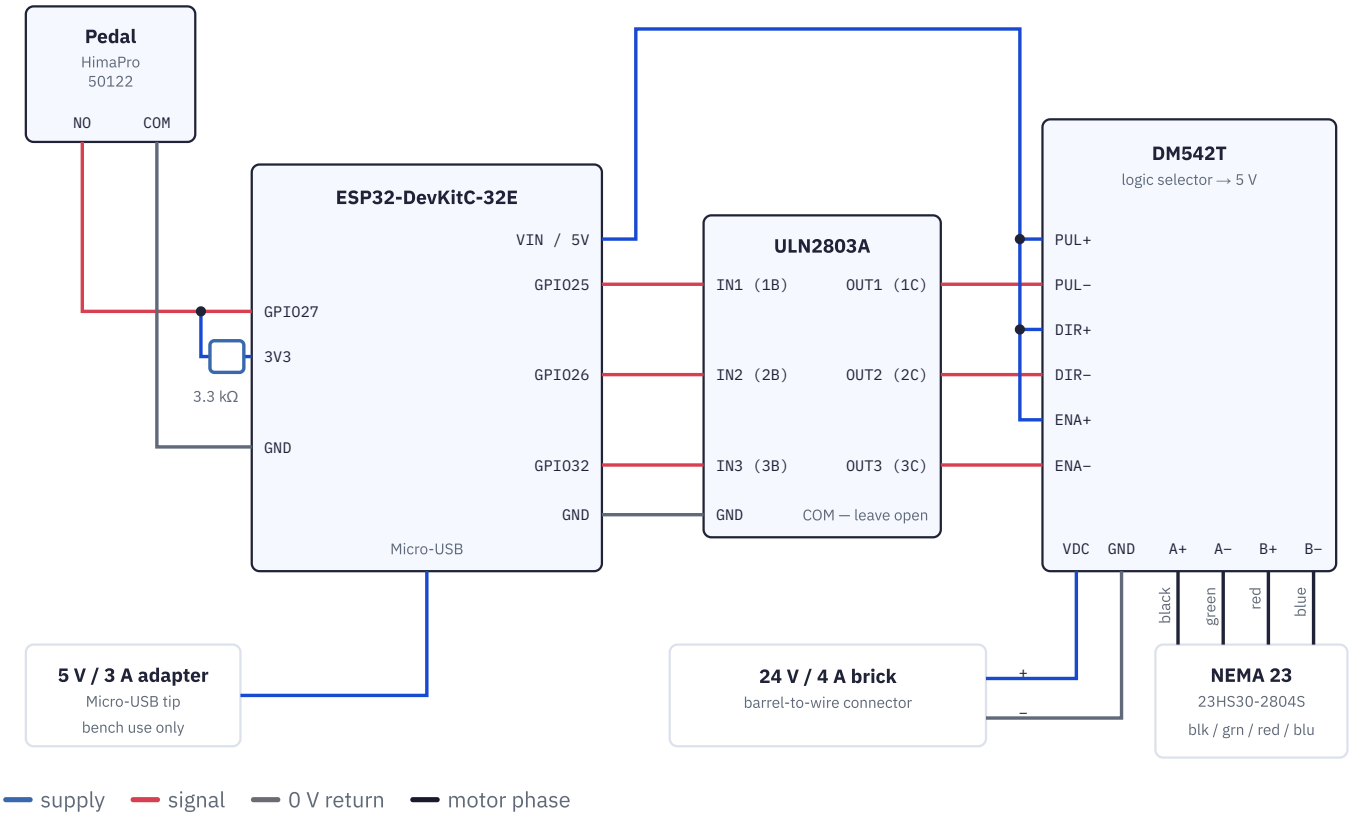
- Find the **signal-voltage selector** and set it to **5 V**. It is separate from the eight DIP switches. Get this wrong and the ULN2803A's sunk outputs will not register as pulses.
- ENA+ and ENA– are wired on the next page. Through them the ESP32 lets the motor go cold ten seconds after each release.

3,200 IS LOAD-BEARING

The firmware's speeds and its degrees readout assume it: 3,200 × 4.5 = **14,400 pulses per table revolution**. Change SW6–SW8 and every speed and every reading is wrong.

The wiring

Every conductor in the controller, once. Ferrule the ends, sleeve the pedal lead in 1/4 inch braid, and leave both supplies unplugged until page 36 passes.



Making each joint

1. Strip 22 AWG to the ferrule's barrel length, crimp a ferrule on every end that lands in a screw terminal, and heat-shrink the transition.
2. Twist the pedal's `COM` and `NO` conductors together along their whole run and sleeve them in 1/4 inch braid.
3. Fit the 3.3 kΩ resistor at the **controller** end, not the pedal end.
4. Leave the 5 V adapter and the 24 V brick unplugged. They do not go in until page 36 has passed and page 39 says so.

A BROKEN PEDAL WIRE READS RELEASED

That is the point of the external pull-up. It also drops the input impedance enough to reject the noise a laser welder puts on a long bench lead.

THE PULL-UP IS A BANDED RESISTOR

One from the Chanzon 3.3 kΩ pack: a small blue cylinder banded orange, orange, black, brown, brown. The pedal's WAGO holds `GPIO27`, one of its leads and `NO`, and nothing else; the other lead meets `3V3` in its own WAGO.

Meter it before power

Both supplies still unplugged. Go down this list with the multimeter on continuity and put a mark beside each row.

FROM	TO
ESP32 GPIO25	ULN IN1 (1B)
ULN OUT1 (1C)	DM542T PUL−
ESP32 GPIO26	ULN IN2 (2B)
ULN OUT2 (2C)	DM542T DIR−
ESP32 GPIO32	ULN IN3 (3B)
ULN OUT3 (3C)	DM542T ENA−
ESP32 VIN / 5V	DM542T PUL+, DIR+ and ENA+
ESP32 GND	ULN GND
Pedal COM	ESP32 GND
Pedal NO	ESP32 GPIO27
ESP32 3V3	GPIO27 reads about 3.3 kΩ on the Ω range
Motor black / green	DM542T A+ / A−
Motor red / blue	DM542T B+ / B−
24 V barrel connector + / −	DM542T VDC / GND

Then the four that must *not* beep

DM542T VDC	DM542T GND
DM542T VDC	any ESP32 pin
ESP32 3V3	ESP32 GND — pedal up, then held down
Any motor phase	any other phase's pair

GATE — BEFORE A PLUG GOES IN A SOCKET

Every row in the first table beeps, or reads its 3.3 kΩ, and every row in the second does not. **The 24 V rail must have no path to the ESP32 at all.** The isolation lives in the DM542T's input opto-couplers, and the only way it can be defeated is a wire in the wrong terminal.

ALSO CHECK WITH YOUR EYES

The DM542T's **logic selector is on 5 V**, its eight DIP switches read ON OFF OFF OFF ON ON OFF ON, and the ULN's COM is empty.

CONTROLLER · STEP 26

Plug the ESP32 into the laptop

The board's own Micro-USB, straight to the laptop. Its onboard CP2102 carries the upload, resets the board into the bootloader by itself, and powers it while it is connected.

1. **Unplug the 24 V brick.** If it was ever plugged in, unplug it now. The motor must be dead for everything on this page and the next.
2. Take the Micro-USB **data** cable — the one that has flashed this project's other DevKitC boards, not a charge-only cable.
3. Plug it into the ESP32's own Micro-USB socket. Not the 5 V adapter's tip: that comes back on page 39.
4. Plug the other end into the laptop.
5. Tell me it is connected. I will build and upload `firmware/src_weld_rotator/` to it.

GATE — THE BOARD IS ON THE BUS

A new serial device appears on the laptop. On this Mac it will be a `/dev/cu.usbserial-*` or `/dev/cu.SLAB_USBtoUART`. **No device means a charge-only cable, or the wrong socket.**

NOTHING ABOUT THE UPLOAD OPENS A MOTOR

Building and flashing touches no motor terminal, and the driver has no power. The host's USB is what powers the board through this step and the next.

WHAT LANDS ON IT

- A controller that boots motionless and will not arm until it has seen the pedal released once.
- The pedal is the whole control: held, the table turns at the stored speed; released, it stops.
- A driver that holds the motor while the table turns and for ten seconds after a release, then lets it go cold.
- A console readout of the degrees turned — never a stop.
- Settings stored in NVS, so they survive being unplugged.

DEFAULTS IT ARRIVES WITH

Bead travel	8.0 mm/s
Direction	cw
Console	115200 baud

CONTROLLER · STEP 27

Store the settings

Same cable, same socket. Open the serial console at 115200 baud and type these. They persist in NVS, so you do this once.

```
# the console is serviced only while the table is stopped
status        read back everything it currently holds
direction cw  tube direction, viewed from above
speed 8.0     bead travel, mm/s. 5 to 15 accepted
status        read it back again and check every line
```

The rest of the vocabulary

`direction ccw`	The other way.
`dirinvert on`	Flips which way `cw` means. Issued once, on page 40, if the table turns the wrong way.
`defaults`	Back to 8.0 mm/s and cw.

COMMANDS ARE TAKEN ONLY WHILE STOPPED

Deliberately. Serial formatting and a flash write both take time, and neither may stretch the interval between two weld-step pulses. **The pedal is the live stop control**, not the console.

WHY 8.0

It sits in the middle of the useful window, runs the table at 1.235 rpm, and turns 360° in 48.6 s. Page 47 has the whole table.

GATE — STATUS READS BACK WHAT YOU TYPED

Speed 8.0 and direction `cw`. **If a line did not take, the table was not stopped when you sent it** — send it again.

GATE — HOLD THE PEDAL DOWN

The board's power LED stays lit and the console prints `RUN`. **A light that goes out, or a port that drops, as the pedal goes down means the press reaches 3V3 without the resistor in the way** — unplug, and check the part and its two WAGOs against page 35.

CONTROLLER · STEP 28

Move to bench power

Settings stored. Now the board comes off the laptop and onto the 5 V adapter, and the 24 V brick goes in for the first time.

1. Close the serial console.
2. Unplug the Micro-USB from the laptop end.
3. Fit the **Micro-USB tip** to the universal 5 V / 3 A adapter and plug it into the ESP32's same Micro-USB socket.
4. Plug the 5 V adapter into the wall. The board's power LED comes up.
5. Confirm the pedal is **released** and that nothing — hand, cloth, cable — is inside the belt path or on the turntable.
6. Plug the 24 V brick into the wall. The DM542T's LED comes up, and the table still turns by hand: the driver stays released until the pedal asks for work.

GATE — NOTHING MOVES

With both supplies live and the pedal released, the table stands still and stays still. **Any motion at all here means a pulse is being generated with no pedal** — pull the 24 V and go back to page 36.

STIFF BEFORE THE FIRST PRESS IS A FAULT

An `ENA` left unwired holds the motor whenever the driver has power. Pull the 24 V and meter the `GPIO32`, `3C` and `ENA+` rows on page 36 again. If all three beep, the driver reads `ENA` the other way round from its manual, and that is a firmware change, not a wire.

ORDER MATTERS, BOTH WAYS

5 V first, then 24 V. Coming down, 24 V first, then 5 V. The driver's inputs should never see a floating logic side with its power stage live.

THE CONSOLE IS STILL THERE

The onboard CP2102 drives `RX0/TX0` whenever the board has USB power, so nothing external is ever wired to those pins. To change a setting later, swap the tip back for the laptop cable.

COMMISSIONING · GATE 1

Direction, and the pedal

No tube. Two readings: that `cw` means clockwise from above, and that the pedal is a real deadman.

1. Stick a strip of paper tape on the turntable rim as an index mark.
2. With `direction cw`, tap the pedal and watch the mark **from above**.
3. If it moved counterclockwise, send `dirinvert on` once, then tap again and confirm it is now clockwise.
4. Power the controller down and back up (5 V last off, 5 V first on).
5. **Hold the pedal down while it boots.** The table must not move.
6. Release the pedal. Press it. The table turns.
7. While it is turning, **unplug one pedal lead**. The table must stop immediately.
8. Plug the lead back in and confirm it turns again.
9. Release the pedal and count ten seconds. The motor lets go and the table turns by hand.

GATE — FIVE THINGS, ALL OF THEM

Clockwise from above means clockwise. Held at boot, the table stays still. Released then pressed, it turns. A pulled lead stops it. Ten seconds after a release, it turns by hand. **Do not set the wire's lead/trail orientation, or go near a tube, until the direction convention is true.**

ONCE ONLY

`dirinvert` is issued once and stored. If you find yourself sending it a second time, something else changed — check the motor's A and B pairs before you invert again.

COMMISSIONING · GATE 2

Time three revolutions

Still no tube. This is the reading that says the drive is actually turning at the speed the controller believes it is.

SET	TABLE	ONE REVOLUTION	ACCEPT
speed 5	0.772 rpm	**77.7 s**	76.9–78.5 s
speed 8	1.235 rpm	**48.6 s**	48.1–49.1 s
speed 15	2.316 rpm	**25.9 s**	25.6–26.2 s

1. Set speed 5.
2. Line the index mark up with a fixed point on the base — a corner, a bench hole, anything that does not move.
3. Hold the pedal and start the stopwatch as the mark passes the point. Stop it as the mark comes back round.
4. Repeat at speed 8 and speed 15.
5. Listen while each one runs.

GATE — ±1% ON ALL THREE, AND NO NOISE

Each time lands inside its accept column. **A stall, a skipped tooth, or an audible step loss fails the drive** — and it fails it here, at zero load, which means it will certainly fail under a 2 kg vessel.

A TIME THAT IS LONG, EVENLY, ON ALL THREE

That is a belt slipping teeth, not a motor losing steps. Re-check the tension twist and the tracking on page 27.

The lap

Nothing counts a lap. The table turns while the pedal is held, and a lap — one full turn and about 20° past the first tack — is judged at the index mark. The console's degrees are a readout, never a stop.

1. Send `status` and check the speed.
2. Tap the pedal until the paper index mark lines up with a fixed reference. Mark that reference too, so you have a pair.
3. Press and **hold** the pedal. Keep holding through more than one full turn.
4. Watch the mark come round past the reference and keep going.
5. Release about 20° past the reference, which is about 33 mm round a 190 mm-diameter rim.
6. Read the degrees the console reports at the release and compare them with where the mark stopped.

GATE — ONE STEADY PASS

The table does **not** stop, pause or change speed as the mark comes round, and the console's degrees agree with the mark within one pass. **A table that stops on its own mid-turn fails the drive.**

YOUR FOOT ENDS THE BEAD

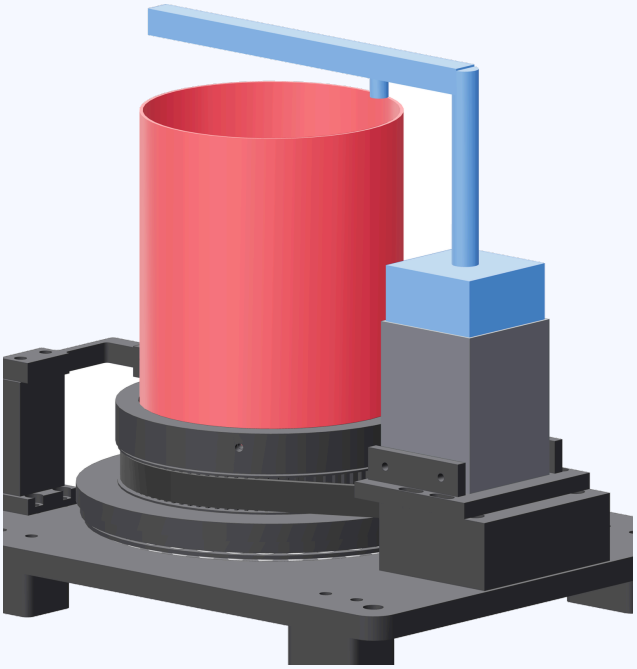
The controller never takes the table away mid-bead. Release the pedal to stop; the console's degrees check the judgement at the index mark and do not make it.

The tube, and the indicator

The first time a tube goes in. Two readings decide whether this fixture is good enough to weld on: radial runout and end-cap face runout.



Straight down, until the cut rim seats. The OD guide starts it, the ID pilot backs it.



The magnetic base on the **motor's exposed steel lamination face** — the only magnetic datum on this fixture.

1. Back all three adjuster tips fully clear. Drop the tube in until its cut rim seats.
2. Bring all three adjusters to **light contact**, then advance them in small equal increments.
3. Seat the indicator's magnetic base on the NEMA 23's bare steel lamination stack — it will not hold on PET-GF or on 316L. Lock both arm joints and **shake-test it**.
4. Put the contact on the tube OD as close to the working end as head clearance allows, turn one slow revolution with the pedal, and read total indicated runout.
5. Adjust the three screws until radial TIR is at or below **0.25 mm**.
6. Seat and tack the end plate per `pressure-vessel.md`, then read the **face** at the weld circle separately.

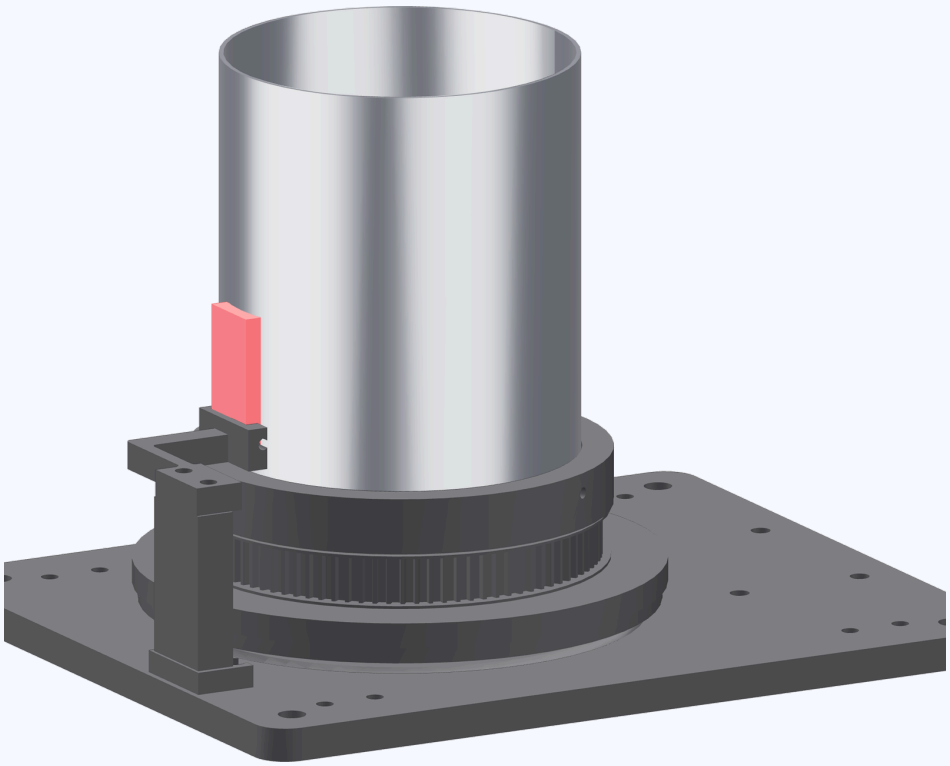
GATE — TWO NUMBERS

Radial runout at the weld end **≤ 0.25 mm TIR**. End-cap face runout at the weld circle **≤ 0.30 mm TIR**. **A part outside either limit is not corrected with more screw force.** Check the lower rim is seated, then square the tube end or correct the end-cap seat.

COMMISSIONING · GATE 5

Continuity, two revolutions

The welder's conductance interlock wants an unbroken path to the work. This is how you prove the shoe gives it one through every angle of the lap.



The loaded tube pushes the shoe back against the fork's rear wall. The shoe wipes the tube 15–65 mm above the nest, and the work cable never moves.

1. **Laser disabled.** Not standby — disabled.
2. Scuff the tube's contact stripe and the copper's contact face. Immediately before use, both times, every time.
3. Clamp the X1 Pro's work lead to the exposed upper 38 mm of the shoe. **Not to the tube, and never to the bearing.**
4. Put the multimeter on continuity between the shoe and the tube.
5. Hold the pedal for two complete revolutions and watch the meter the whole way round.

GATE — NO BLINK

Two complete revolutions with **not one dropout**. **Any blink fails setup.** Clean both faces and take the reading again before the laser is enabled.

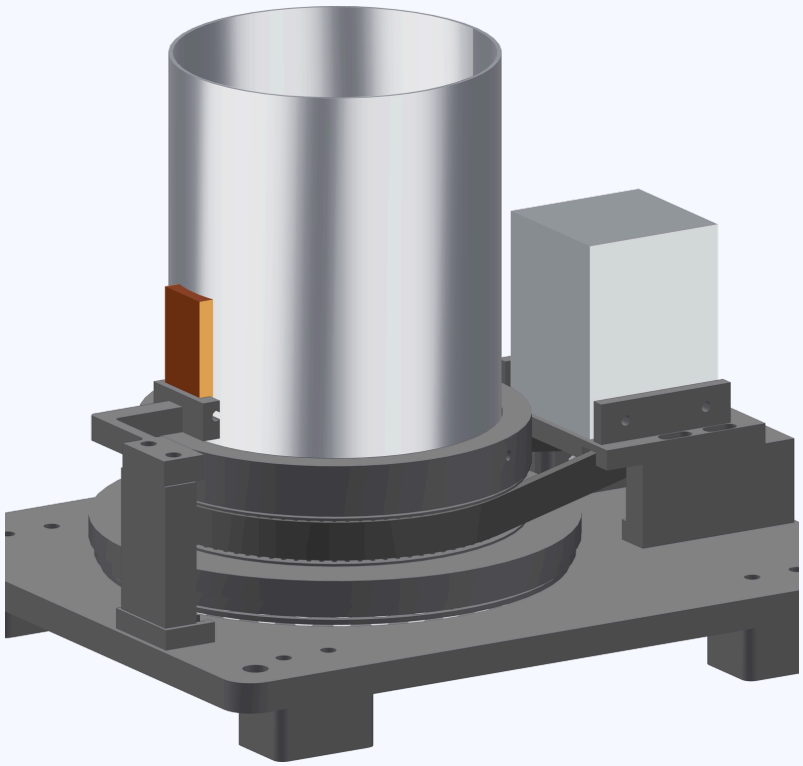
THIS IS NOT A PROTECTIVE EARTH

It exists for the welder's conductance interlock. The PP balls and the printed races are not in the electrical path.

COMMISSIONING · GATE 6

The rehearsal

The last gate, and the only one that involves the whole cell. Everything present, everything in position, laser disabled.



Where you will actually stand. The copper contact is on the near side; the motor and its belt are on the far side.

1. Bring everything to the bench that will be there for a real weld: the head, the wire guide, the purge hose, the work lead, and you in your working position.
2. Confirm the laser is **disabled**.
3. Run one complete lap with the pedal: a full turn and about 20° past the index, ended by releasing it.
4. Watch the belt for the whole lap. Watch the tube for the whole lap.
5. Move your hands and the head through the arc you will actually use while it turns.

GATE — NOTHING ENTERS AND NOTHING WRAPS

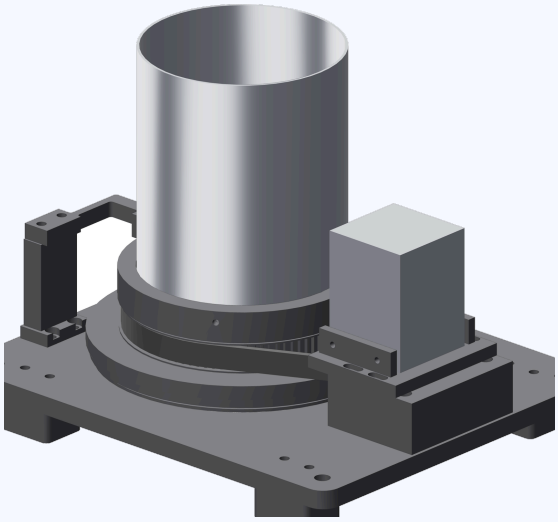
Nothing goes into the belt path. Nothing wraps around the turning tube — not the purge hose, not the work lead, not a sleeve. **Rotation is an additional bench motion and it gets its own pinch-point clearance before the laser is enabled.**

NOTHING ON THIS PAGE RELAXES ANYTHING

Laser guarding, eye protection, extraction, argon-asphyxiation control and pressure-vessel inspection all stand exactly as they did. This gate is in addition to them.

The per-weld sequence

Every closure, in this order. This is the page to leave open on the bench.



1. Clamp the base to the bench through its four 10 mm holes. Confirm the laser is disabled and the pedal released.
2. Seat the tube's *opposite* end in the nest. Bring all three adjusters to light contact and indicate the working end to **≤ 0.25 mm** radial TIR.
3. Seat and tack the end plate per `pressure-vessel.md`. Verify **≤ 0.30 mm** face TIR. **First closure:** back-purge argon through the open held end. **Second closure:** purge into one completed lower port through the raised Ø90 mm passage, other lower port open as the vent.
4. Scuff and engage the copper shoe. Prove uninterrupted continuity for one dry revolution.
5. While stopped, set the speed. Return the index to the first tack. Establish shielding and place the wire on the **arriving** side of the puddle for the verified direction.
6. Hold the head at the qualified angle and standoff. **Press and hold the pedal.** Once rotation is steady, hold the laser trigger for the bead. Carry the bead about 20° past the first tack, judged at the index mark with the console's degrees as a check, and stop there.
7. Release the trigger, then the pedal. A wire that sticks in the puddle is **snipped, not lasered off**: keep the head where it stopped and cut the wire between the wire nozzle and the bead.
8. Continue with PT and hydrostatic inspection in `pressure-vessel.md`. Record speed, direction, the degrees the console reported at release, and runout beside the result.

The numbers

Travel speed is the recipe setting; table rpm follows from it. The controller clamps every command into the 5–15 mm/s window — a commissioning range, not a qualified weld schedule.

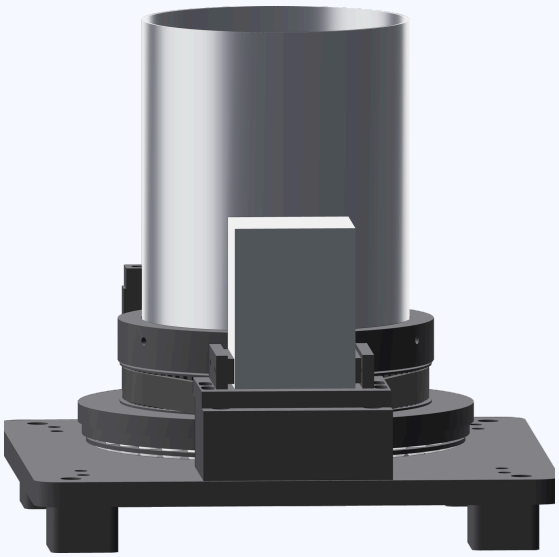
TRAVEL	TABLE	360°	PULSES/S
5 mm/s	0.772 rpm	77.7 s	185.3
6 mm/s	0.926 rpm	64.8 s	222.3
8 mm/s	**1.235 rpm**	**48.6 s**	**296.4**
10 mm/s	1.544 rpm	38.9 s	370.6
12 mm/s	1.853 rpm	32.4 s	444.7
15 mm/s	2.316 rpm	25.9 s	555.8

The joint

End plate	ID-fit plug, 6.35 mm recess
Weld-circle diameter	123.70 mm
One revolution at the bead	388.61 mm
Rotating mass, 1st / 2nd closure	1.40 / 2.01 kg

The drive

Reduction	4.5:1 — 20T to 90T
Driver microstepping	3,200 pulses/motor rev
Table resolution	14,400 pulses/rev
One pulse	0.025°, or 0.027 mm at the bead
Overlap past the first tack	about 20°, at the index mark
Motor speed at 8 mm/s	5.558 rpm
Belt centre distance	125.1 mm nominal



The tube's working rim stands **238.4 mm** above the bench. The highest PET-GF anywhere near it is the ground arm at 91 mm, so there is more than 125 mm of metal between the closure weld and any printed structure.

Start at 8 and change one variable at a time after the first 316L coupon. Coupon evidence sets the production value.

WHERE EVERYTHING LIVES

The rest of it

This guide is the build. These are the files it stands on, and where to go when a number here is not enough.

FOR	READ
The fixture, subsystem by subsystem	`hardware/assembly/weld-rotation-rig.md`
Printable geometry, STEP files, print notes	`hardware/printed-parts/fixtures/weld-rotator/README.md`
The controller's own reference	`firmware/src_weld_rotator/README.md`
What the fixture is tooling for	`hardware/assembly/pressure-vessel.md`
Every part, order and status	`hardware/ledger/purchases.md`
The print profile's first layer	`hardware/printed-parts/z-trim.md`
Archived illustration source	`rotator_art.py`
This document	`hardware/weld-rotator-guide/README.md`

Regenerating the fixture

The fixture README gives the native geometry build and its deliberate fit checks.

Rebuilding the controller

```
~/.platformio/penv/bin/pio test -e native -f test_weld_rotator_policy
~/.platformio/penv/bin/pio run -e weld_rotator
```

THIS DOCUMENT IS BUILT BY HAND

Edit the numbered HTML leaves and committed illustrations, then run `tools/run hsm-manual-guides bind weld-rotator-guide`. The command renders and binds this guide on demand, outside the appliance build.